

Contrasting conservation conditions shape the spatial partitioning of zooplankton diversity in Atlantic Forest streams

Contrasting conservation conditions shape the spatial partitioning of zooplankton diversity in Atlantic Forest streams ON Acta Limnologica Brasiliensia

Ellen Gomes da Silva Mayara Mirelly da Silva Monteiro
Marcela Vitória Bernardo da Silva
Ludmilla Cavalcanti Antunes Lucena Thais Xavier de Melo
Elvio Sergio Figueredo Medeiros

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Abstract

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1 Introduction

Essas abordagens são baseadas nos conceitos de turnover and nestedness. Species turnover and nestedness are two distinct components used to break down and understand beta diversity—which measures how much species composition changes between different communities or sites (Baselga 2010), where, turnover is the replacement of some species by different species from

one site to another, while nestedness is the loss or gain of species where poorer sites contain a strict subset of the species found at richer sites.

Understanding the spatial structure and distribution of biodiversity across scale remains a central theme in community ecology and conservation planning (Lande 1996, Crist et al. 2003). Total regional species richness (γ -diversity) can be hierarchically partitioned into its additive components: local richness (α -diversity) and species turnover among assemblages (β -diversity) across spatial or temporal scales (Lande 1996). Examining these components allows ecologists to disentangle whether regional diversity is primarily governed by fine-scale local processes or broader spatial heterogeneity (Crist et al. 2003, Tuomisto 2010). Beta diversity (β) itself reflects the variation in species composition among sites and is driven by two distinct ecological phenomena: turnover (species replacement) and nestedness (richness differences) (Baselga 2010). Species turnover occurs when species are replaced by different species along environmental or spatial gradients, usually driven by environmental filtering, competitive exclusion, or dispersal limitation (Baselga 2010, Tuomisto 2010). In contrast, nestedness occurs when communities at species-poor sites are non-random subsets of those at species-rich sites (Baselga 2010). Nested patterns typically stem from deterministic processes such as selective extinction, differential colonization capacity, or habitat fragmentation that sequentially eliminate sensitive taxa while retaining tolerant generalists (Baselga 2010, Gotelli & Ulrich 2012). Evaluating the relative contribution of turnover and nestedness—alongside null models using Standardized Effect Sizes (SES)—provides critical insights into assembly rules and conservation prioritization (Gotelli & McCabe 2002, Baselga 2010). A pattern dominated by turnover ($SES > 0$) implies strong ecological differentiation and high beta diversity, signaling that distinct localities host unique biological assemblages (Baselga 2010). From a conservation perspective, high turnover necessitates a network-wide preservation strategy, as multiple sites must be protected to encompass regional biodiversity (Baselga 2010). Conversely, a system dominated by nestedness or fine-scale homogenization ($SES < 0$) indicates that local assemblages are nested sub-collections of regional species pools. Under a nested regime, management efforts are most effectively concentrated on protecting the largest or highest-quality species-rich sites, which inherently harbor the vast majority of the regional pool (Baselga 2010).

CITAR: (Gering et al. 2003, Crist et al. 2003, Veech et al. 2003, Ligeiro et al. 2010, Baselga 2010, 2012, Ávila et al. 2011, Shell and Collier 2018)

1.1 Objectives

This study describes the richness and density of zooplankton, comparatively between two small head water streams subject to different levels of environmental pressures. One originates within a protected area - Biological Reserve, or REBIO, according to Brasil (2000) - and its course extends beyond its administrative boundaries (MMA 2003), and a second one that flows entirely outside of the protected area. Within this framework, we use hierarchical additive partitioning to evaluate the relative contribution of different spatial and temporal scales to the diversity of zooplankton assemblages. Given the importance of the Atlantic Forest to the

biodiversity (Pereira et al. 2026) and its fragmented stated north of the São Francisco river basin (Silva and Casteleti 2003), we aim with this study, (1) to infer environmental degradation from species turnover (Baselga 2010) and (2) to associate beta-diversity to the ecological health of sites inside and surrounding a protected area of Atlantic Forest. Given the expected different levels of environmental pressures in the area (Gouveia et al. 2017), we hypothesize that the diversity is not be homogeneously distributed among the different spatial scales but concentrated at the finer levels and seasonal variation will be limited to species composition, but no spacial variation is expected between dry and wet seasons. Some environmental features (reach morphology, habitat structure, and water quality) are also described to explore possible explaining factors for the observed patterns.

2 Material and methods

2.1 Study area and sampling design

This study was conducted in the Guaribas Biological Reserve (REBIO Guaribas), a protected area of Atlantic Forest in northeastern Brazil. This protected area is located in a transitional zone between the Caatinga and Atlantic Forest ecoregions. The land use of the study area is based on the occupation of traditional human populations and an increasing use of land for large scale monocultures, mostly sugar cane (Heinrichs et al. 2017). Other than these, small scale agricultural activities, fishing, and the modification of the natural vegetation for livestock are important anthropogenic pressures that affect the buffer zone of the REBIO (Arruda et al. 2013, MMA 2026). Other common uses of the soil in the area are, coconut plantations, orchards and subsistence agriculture. These land uses represent important sources of pollution, either through agrochemicals runoff or direct and indirect domestic effluents (such as septic tanks). These nonpoint source pollutants contaminate underground water and create a spatial gradient of environmental disturbance (Mehla et al. 2026). Therefore, the mainly rural nature of the land use associated with the management of vegetation, will cause direct impact to aquatic systems, by the siltation of streams and contamination of groundwater. The hydrographic network of the study region is composed predominantly of small catchments containing first- and second-order streams (Strahler 1957 classification) (Figure 1). These watercourses are tributaries of the Camaratuba River and are sustained by numerous springs distributed throughout the region. The two main stream systems in the study area are the Barro Branco and Caiana catchemnts. The Barro Branco stream is approximately 8.8 km long and originates within the protected area. Its upper section remains within the boundaries of the REBIO, whereas its middle and lower sections extend into the surrounding landscape. Despite this transition, riparian vegetation is preserved along most of the stream. The Caiana stream is approximately 9.8 km long and also originates within the REBIO, although most of its course lies outside the protection area.

The climate in the region is classified as As' (Peel et al. 2007), characterized by a mean annual temperature of approximately 25 °C, with maximum temperatures occurring mainly between

February and March, during the dry period, as well as by low annual thermal amplitude. Annual precipitation ranges from 1,400 to 1,800 mm, concentrated between March and July, whereas the lowest rainfall levels occur between September and December. In the study area, rainfall is concentrated between March and July, beginning in autumn and extending into winter. This pattern is attributed to the influence of cold air masses from the South Atlantic ocean that affect the region. The hydrographic network is composed of small headwater streams fed by rain (Gouveia et al. 2017).

A total of 15 sampling sites distributed longitudinally across the two Atlantic Forest streams were selected to capture spatial variation in environmental conditions and habitat characteristics. The Barro Branco stream included sampling sites located within a protected area (REBIO Guaribas) and sites located outside its boundaries, whereas all sampling sites in the Caiana stream were located outside the protected area. This sampling design allowed the assessment of zooplankton diversity across nested spatial scales and contrasting conditions of environmental protection and human influence (Figure 1). Sampling included temporal (dry and wet seasons) and spatial scales. The spatial scales of zooplankton diversity were represented by the total gamma (γ) diversity, encompassing both study streams. At each stream, several sites were selected according to their position in the main channel of the catchment basin and, at each site, three habitats were sampled to represent different habitat types (pool, riffle, near underwater structures) (Frissell et al. 1986). We also considered a further spatial scale being the upper reaches (portion of the Barro Branco stream within the conservation unit and the Caiana stream before its confluence with the Pipina stream) and the lower reaches (portion of the Barro Branco stream outside the conservation unit and the Caiana stream after its confluence with the Pipina stream) (Figure 1). Sampling was conducted during the dry (January-February) and wet (June-July) seasons of 2024.

2.2 Environmental data collection

Environmental conditions were characterized using four groups of variables, following procedures adapted by Medeiros et al. (2008) from the protocols proposed by Pusey et al. (2004) and Mugodo et al. (2006): physicochemical water quality, stream site morphology, substrate composition, and marginal habitat structure. Water quality was characterized using physical and chemical measurements obtained with portable instruments, including water temperature ($^{\circ}\text{C}$) and dissolved oxygen concentration (mg/L). Turbidity was subsequently measured in the laboratory with a turbidimeter using water samples collected at each sampling site. Current velocity (m/s) was estimated using the float method, as described by Maitland (1990). Morphological characteristics were assessed from three randomly selected transects at each site, where stream width (cm) and water depth (cm) were measured. Bank slope was classified into four categories according to its inclination: 1, $< 20^{\circ}$; 2, $> 20-30^{\circ}$; 3, $> 30-50^{\circ}$; and 4, $> 50^{\circ}$. Broader-scale landscape characteristics, including elevation and stream length, were obtained from handheld GPS records and satellite imagery. Substrate composition and the physical structure of marginal habitats were evaluated from 9 to 12 quadrats (1 m^2) distributed along

the terrestrial-aquatic interface at each sampling site. Within each quadrat, the proportional cover of different substrate categories was visually estimated, including mud, sand, cobbles, fine and coarse gravel and rocks. The same quadrats were used to characterize littoral and submerged habitat features. Recorded habitat components included macrophytes, marginal grasses, leaf litter, attached and filamentous algae, overhanging and submerged vegetation, small and large woody debris, and root masses (Medeiros et al. 2008). For subsequent analyses and presentation, environmental measurements were summarized as mean values for each sampling site.

Three 250 mL replicate samples of water from each site were taken and refrigerated for concentration analysis (g L^{-1}) of ammonia, nitrite, nitrate, orthophosphate (soluble reactive phosphorus) and total phosphorus. Nutrient concentrations were obtained based on Standard Methods for the Examination of Water (1998) and Mackereth et al. (1978).

2.3 Zooplankton sampling

Zooplankton was sampled at habitats within each sampling site, hereafter referred to as habitat samples. Collections were performed in the main channel of the streams at the surface water. At each sampling habitat, a plankton net (30 cm mouth diameter, 70 cm length, and 60 μm mesh size) was used to filter 50 L of water. The net was rinsed between successive samples to prevent clogging and carryover of organisms. Each sample was concentrated to a final volume of 80 mL for laboratory analyses. Following collection, zooplankton samples were anesthetized with commercial sparkling water and subsequently preserved in 4% formalin (EPA 2020). Sucrose was added to the preserved sample to prevent female cladocerans from losing eggs and to minimize carapace distortion (Haney and Hall 1973). In the laboratory, samples had their zooplankton identified and counted using a Sedgewick–Rafter counting chamber. The analyses focused on representatives of Rotifera, Cladocera, and Copepoda. Taxonomic identification was based on specialized literature and confirmed by specialists. To evaluate whether the sampling effort adequately represented the zooplankton assemblage, species accumulation curves were calculated (Melo et al. 2014, Lucena et al. 2015).

2.4 Statistical analyses

Environmental variables were screened for multicollinearity and to identify redundant descriptors. Variables representing similar environmental information were also combined when appropriate, thereby reducing redundancy in the dataset (Legendre and Legendre 1998).

```

1 #ZOOREBIO23_PART - ORGANIZANDO DADOS AMBIENTAIS----
2 #####....-----
3
4 dev.off() #apaga os graficos, se houver algum
5 rm(list=ls(all=TRUE)) #limpa a memória

```

```

6  cat("\014") #limpa o console
7
8  ##CARREGANDO MATRIZES BRUTAS-----
9
10 library(openxlsx)
11 habitat <- read.xlsx("dados/rebio23-habitat.xlsx",
12                     rowNames = T,
13                     colNames = T,
14                     sheet = "ambiente",
15                     startRow = 2)
16
17 # List columns with at least one NA
18 colnames(habitat)[colSums(is.na(habitat)) > 0]
19 # Identify columns containing only NA values
20 all.na <- names(habitat)[colSums(!is.na(habitat)) == 0]
21 all.na <- c(all.na, "w.Temp_C", "w.DO_mg.L")
22 #all.na <- c(all.na, "n.Ammonia_NH3_ug.L", "n.Nitrite_NO2_ug.L",
23 #            "n.Nitrate_NO3_ug.L", "n.Orthophosphate_SRP_ug.L",
24 #            "n.Phosphorus_TP_ug.L")
25 # Remove those columns by name
26 m_hab_part <- habitat[, !names(habitat) %in% all.na, drop = FALSE]
27 # Removendo linhas por nome
28 rownames(m_hab_part)
29 del_rows <- c("C-C-P11-H1", "C-C-P11-H2", "C-C-P11-H3",
30              "S-C-P11-H1", "S-C-P11-H2", "S-C-P11-H3")
31 del_rows
32 m_hab_part <- m_hab_part[!(row.names(m_hab_part) %in% c(del_rows)),]
33 m_hab_part
34 # List columns with at least one NA
35 colnames(m_hab_part)[colSums(is.na(m_hab_part)) > 0]
36 #rownames(m_hab_part)[is.na(m_hab_part$m.Width_m)]
37 # Remove columns with at least one NA
38 m_hab_part <- m_hab_part[, colSums(is.na(m_hab_part)) == 0, drop = FALSE]
39
40 ###REMOVENDO COLUNAS ZERADAS DE HABITAT----
41
42 sum <- colSums(m_hab_part)
43 sum
44 zero_sum <- names(which(colSums(m_hab_part) == 0))
45 zero_sum #nomes das colunas zeradas
46 m_part_cols <- m_hab_part[(colSums(m_hab_part) != 0)] #em != a exclamação inverte o sentido
47 zero_sum2 <- names(which(colSums(m_part_cols) == 0))
48 zero_sum2 #nomes das colunas zeradas

```

```

49 sum<-colSums(m_part_cols)
50 sum
51
52 t_grps <- read.xlsx("dados/rebio23-habitat.xlsx",
53                     rowNames = T,
54                     colNames = T,
55                     sheet = "grupos",
56                     startRow = 2)
57
58 ###REMOVENDO LINHAS ZERADAS DE GRUPOS----
59
60 rownames(t_grps)
61 del_rows
62 t_grps <- t_grps[!(row.names(t_grps) %in% c(del_rows)),]
63 t_grps
64
65 ###SEPARANDO NUTRIENTES----
66
67 nutri <- grep("^n\\.\\.", colnames(habitat), value = TRUE)
68 nutri
69 nutrientes <- habitat[, nutri, drop = FALSE]
70 nutrientes <- nutrientes[, colSums(!is.na(nutrientes)) > 0, drop = FALSE]
71 nutrientes <- na.omit(nutrientes)
72
73 ###SALVANDO MATRIZES FINAIS----
74
75 write.table(m_part_cols, "res/m_hab_part.csv",
76             sep = ";", dec = ".", #"\t",
77             row.names = TRUE,
78             quote = TRUE,
79             append = FALSE)
80 write.table(nutrientes, "res/t_nutri.csv",
81             sep = ";", dec = ".", #"\t",
82             row.names = TRUE,
83             quote = TRUE,
84             append = FALSE)
85 write.table(t_grps, "res/t_grps.csv",
86             sep = ";", dec = ".", #"\t",
87             row.names = TRUE,
88             quote = TRUE,
89             append = FALSE)
90 t_grps <- read.csv("res/t_grps.csv",
91                   sep = ";", dec = ".",

```

```

102         row.names = 1,
103         header = TRUE,
104         na.strings = NA)
105 t_nutri <- read.csv("res/t_nutri.csv",
106         sep = ";", dec = ".",
107         row.names = 1,
108         header = TRUE,
109         na.strings = NA)
110 m_hab <- read.csv("res/m_hab_part.csv",
111         sep = ";", dec = ".",
112         row.names = 1,
113         header = TRUE,
114         na.strings = NA)
115
116 #ZOOEBIO23_PART - ORGANIZANDO DADOS DA COMUNIDADE-----
117 #####....-----
118
119 dev.off() #apaga os graficos, se houver algum
120 rm(list=ls(all=TRUE)) #limpa a memória
121 cat("\014") #limpa o console
122
123 library(openxlsx)
124
125 ##CARREGANDO MATRIZES BRUTAS-----
126
127 densidade <- read.xlsx("dados/rebio23-zoopl.xlsx",
128         rowNames = T,
129         colNames = T,
130         sheet = "densidade_hab_indL")
131 densidade[1:5,1:5] #[1:5,1:5] mostra apenas as linhas e colunas de 1 a 5.
132
133 ##COLUNAS ZERADAS DE DENSIDADE-----
134 sum <- colSums(densidade)
135 sum
136 zero_sum <- names(which(colSums(densidade) == 0))
137 zero_sum #nomes das espécies zeradas
138 m_part_cols <- densidade[(colSums(densidade) != 0)] #em != a exclamação inverte o sentido
139 zero_sum2 <- names(which(colSums(m_part_cols) == 0))
140 zero_sum2 #nomes das espécies zeradas
141 sum<-colSums(m_part_cols)
142 sum
143
144 ##LINHAS ZERADAS DE DENSIDADE-----

```

```

135 m_part2 <- as.data.frame(t(m_part_cols))
136 sum <- colSums(m_part2)
137 sum
138 zero_sum <- names(which(colSums(m_part2) == 0))
139 zero_sum #nomes das espécies zeradas
140 m_part_rows <- m_part2[(colSums(m_part2) != 0)] #em != a exclamação inverte o sentido
141 zero_sum2 <- names(which(colSums(m_part_rows) == 0))
142 zero_sum2 #nomes das espécies zeradas
143 sum<-colSums(m_part_rows)
144 sum
145 m_dens <- as.data.frame((t(m_part_rows)))
146
147 ##SALVANDO MATRIZES FINAIS REMOVIDAS AS LINHAS/COLUNAS ZERADAS----
148
149 write.table(m_dens, "res/m_dens.csv",
150             sep = ";", dec = ".", #"\\t",
151             row.names = TRUE,
152             quote = TRUE,
153             append = FALSE)
154 m_dens <- read.csv(res/"m_dens.csv",
155                  sep = ";", dec = ".",
156                  row.names = 1,
157                  header = TRUE,
158                  na.strings = NA)
159
160 #ZOOEBIO23_PART - MATRIZ DE CONTAGEM----
161 #####....----
162
163 library(openxlsx)
164 contagem <- read.xlsx("dados/rebio23-zoopl.xlsx",
165                      rowNames = T,
166                      colNames = T,
167                      sheet = "rebio23_80ml",
168                      startRow = 4)
169 contagem[1:5,1:5] #[1:5,1:5] mostra apenas as linhas e colunas de 1 a 5.
170
171 ###COLUNAS ZERADAS DE CONTAGEM----
172 sum <- colSums(contagem)
173 sum
174 zero_sum <- names(which(colSums(contagem) == 0))
175 zero_sum #nomes das espécies zeradas
176 m_part_cols <- contagem[(colSums(contagem) != 0)] #em != a exclamação inverte o sentido
177 zero_sum2 <- names(which(colSums(m_part_cols) == 0))

```

```

178 zero_sum2 #nomes das espécies zeradas
179 sum<-colSums(m_part_cols)
180 sum
181
182 ###LINHAS ZERADAS DE CONTAGEM----
183 m_part2 <- as.data.frame(t(m_part_cols))
184 sum <- colSums(m_part2)
185 sum
186 zero_sum <- names(which(colSums(m_part2) == 0))
187 zero_sum #nomes das espécies zeradas
188 m_part_rows <- m_part2[(colSums(m_part2) != 0)] #em != a exclamação inverte o sentido
189 zero_sum2 <- names(which(colSums(m_part_rows) == 0))
190 zero_sum2 #nomes das espécies zeradas
191 sum<-colSums(m_part_rows)
192 sum
193 m_cont <- (t(m_part_rows))
194 m_cont
195 sum(m_cont)
196 ncol(m_cont)
197
198 ##SALVANDO MATRIZ FINAL DE CONTAGEM----
199
200 write.table(m_cont, "res/m_cont.csv",
201             sep = ";", dec = ".", #"\\t",
202             row.names = TRUE,
203             quote = TRUE,
204             append = FALSE)
205 m_cont <- read.csv("res/m_cont.csv",
206                   sep = ";", dec = ".",
207                   row.names = 1,
208                   header = TRUE,
209                   na.strings = NA)
210 m_cont
211
212 #ZOOEBIO23_PART - ORGANIZANDO ESCALAS DA COMUNIDADE----
213 #####.....----
214
215 dev.off() #apaga os graficos, se houver algum
216 rm(list=ls(all=TRUE)) #limpa a memória
217 cat("\\014") #limpa o console
218
219 library(openxlsx)
220

```

```

221  ##CARREGANDO MATRIZES BRUTAS-----
222
223  escalas <- read.xlsx("dados/rebio23-zoopl.xlsx",
224                      rowNames = T,
225                      colNames = T,
226                      sheet = "escalas_final",
227                      startRow = 2)
228
229  rownames(escalas)
230  del_rows <- c("C-C-P11-H1", "C-C-P11-H2", "C-C-P11-H3", "S-C-P11-H1", "S-C-P11-H2", "S-C-P11-H3")
231  del_rows
232  escalas <- escalas[!(row.names(escalas) %in% c(del_rows)),]
233
234  write.table(escalas, "res/m_esc.csv",
235             sep = ";", dec = ".", #"\\t",
236             row.names = TRUE,
237             quote = TRUE,
238             append = FALSE)
239  m_esc <- read.csv("res/m_esc.csv",
240                  sep = ";", dec = ".",
241                  row.names = 1,
242                  header = TRUE,
243                  na.strings = NA)
244  m_esc

```



```

1  #REBIO23-part----
2
3  ##ORGANIZANDO DADOS-----
4
5  dev.off()
6  rm(list=ls(all=TRUE))
7  cat("\\014")
8
9  t_grps <- read.csv("res/t_grps.csv",
10                  sep = ";", dec = ".",
11                  row.names = 1,
12                  header = TRUE,
13                  na.strings = NA)
14  m_hab <- read.csv("res/m_hab_part.csv",
15                  sep = ";", dec = ".",
16                  row.names = 1,
17                  header = TRUE,
18                  na.strings = NA)

```

```

19
20 ##CORRELOGRAMA E REMOÇÃO DE VARIÁVEIS REDUNDANTES OU DESNECESSÁRIAS----
21
22 library(psych)
23 colnames(m_hab)
24 #png("res/fig-h.hab_pairs.png")
25 pairs.panels(m_hab[,1:8],
26             method = "pearson", # correlation method
27             scale = FALSE, lm = FALSE,
28             hist.col = "#00AFBB", pch = 19,
29             density = TRUE, # show density plots
30             ellipses = TRUE, # show correlation ellipses
31             alpha = 0.5)
32 #dev.off()
33
34 cor <- cor(m_hab)
35 cor
36
37 library(corrplot)
38 #png("res/fig-hab_corrplot.png")
39 corrplot(cor, method = "circle")
40 #dev.off()
41
42 ##DELETANDO COLINEARES OU INDESEJADAS----
43
44 #sink(file = "res/colineares.txt", append = F, split = T)
45 colnames(m_hab)
46 del_cols <- c("", "")
47 #del_cols <- grep("~n\\. ", colnames(m_hab), value = TRUE)
48 m_hab_part <- m_hab[, !(colnames(m_hab) %in% del_cols)]
49
50 ##SOMANDO REDUNDANTES----
51
52 m_hab_part$h.Algae <- m_hab_part$h.Algae_f + m_hab_part$h.Algae_a
53 m_hab_part <- m_hab_part[, !(colnames(m_hab_part)
54                             %in% c("h.Algae_f", "h.Algae_a"))]
55 m_hab_part$h.Debris <- m_hab_part$h.Debris_l + m_hab_part$h.Debris_s
56 m_hab_part <- m_hab_part[, !(colnames(m_hab_part)
57                             %in% c("h.Debris_l", "h.Debris_s"))]
58
59 colnames(m_hab_part)
60 m_hab_part
61 #sink()

```

```

62
63 write.table(m_hab_part, "res/m_hab_part.csv",
64             sep = ";", dec = ".", #"\t",
65             row.names = TRUE,
66             quote = TRUE,
67             append = FALSE)
68 m_hab_part <- read.csv("res/m_hab_part.csv",
69                       sep = ";", dec = ".",
70                       row.names = 1,
71                       header = TRUE,
72                       na.strings = NA)

1  colnames(m_hab_part)
2  library(PerformanceAnalytics)
3  chart.Correlation(m_hab_part[, c(1:20)], histogram=TRUE)
4  # ***p<0.001, **p<0.01, *p<0.05, .p<0.10
5
6  library(Hmisc)
7  mat <- as.matrix(m_hab_part[, 1:20])
8  res <- rcorr(mat, type = "pearson")
9  cor_mat <- res$r
10 p_mat <- res$P
11
12 stars <- ifelse(p_mat < 0.001, "***",
13               ifelse(p_mat < 0.01, "**",
14                     ifelse(p_mat < 0.05, "*", "")))
15
16 cor_star <- ifelse(p_mat < 0.05,
17                   paste0(round(cor_mat, 2), stars),
18                   NA)
19
20 cor_star <- matrix(
21   cor_star,
22   nrow = nrow(cor_mat),
23   dimnames = dimnames(cor_mat)
24 )
25
26 cor_star[upper.tri(cor_star, diag = TRUE)] <- NA
27 cor_star <- noquote(cor_star)
28 print(cor_star, na.print = "")

1  #PCA fviz package----
2  #browseURL("https://www.sthda.com/english/articles/31-principal-component-methods-in-r-pract

```

```

3
4  #ORGANIZANDO DADOS----
5
6  dev.off()
7  rm(list=ls(all=TRUE))
8  cat("\014")
9
10 t_grps <- read.csv("res/t_grps.csv",
11                   sep = ";", dec = ".",
12                   row.names = 1,
13                   header = TRUE,
14                   na.strings = NA)
15 m_hab_part <- read.csv("res/m_hab_part.csv",
16                      sep = ";", dec = ".",
17                      row.names = 1,
18                      header = TRUE,
19                      na.strings = NA)
20
21 colnames(m_hab_part)
22
23 ###RELATIVIZAÇÕES E TRANSFORMAÇÕES----
24 library(vegan)
25 library(tidyverse)
26
27 # Selecionando variáveis que começam com "w" ou "m"
28 m_wm <- m_hab_part[, grep("(w|m)", colnames(m_hab_part))]
29
30 # Selecionando variáveis que começam com "s" ou "h"
31 m_sh <- m_hab_part[, grep("(s|h)", colnames(m_hab_part))]
32
33 # Removendo os dois primeiros caracteres dos nomes das colunas
34 colnames(m_wm) <- sub("^..", "", colnames(m_wm))
35 colnames(m_sh) <- sub("^..", "", colnames(m_sh))
36
37 # Transformações
38 m_wm <- sqrt(m_wm)
39 m_sh <- asin(sqrt(decostand(m_sh, method = "total", MARGIN = 2)))
40
41 # Juntando tudo em uma única tabela
42 hab_trns <- cbind(m_sh, m_wm)
43
44 # Salvado m_hab_trns
45 write.table(hab_trns, "res/m_hab_trns.csv",

```

```

46     sep = ";", dec = ".", #"\t",
47     row.names = TRUE,
48     quote = TRUE,
49     append = FALSE)
50 m_hab_trns <- read.table("res/m_hab_trns.csv",
51                          sep = ";", dec = ".",
52                          row.names = 1,
53                          header = TRUE,
54                          na.strings = NA)

```

These data were then analyzed using Principal Component Analysis (PCA) to summarize multivariate patterns of environmental variation among sampling sites. Prior to PCA, site morphology and water-quality variables were square-root transformed (Sokal and Rohlf 1995). Variables describing sediment composition and marginal habitat structure, expressed as percentages, were relativized by their column totals and subsequently subjected to the arcsine square-root transformation (McCune and Grace 2002). PCA was conducted in R using the **FactoMineR** package (Lê et al. 2008), with all variables centered and standardized to unit variance before analysis. The water quality parameters temperature and dissolved oxygen, and all nutrients were used to characterize differences between streams but were not used in the PCA because they were not measured for both seasons. A Permutational Multivariate Analysis of Variance (PERMANOVA, 999 permutations) was performed to test whether the overall set of environmental variables measured differed across the seasons and streams .

To compare the diversity of zooplankton taxa between streams, the Chao1 estimated richness ($\pm$ SE) and the curves of rarefied and extrapolated richness were generated (95% upper and lower confidence intervals) using individual-based rarefaction curves (Gotelli and Colwell 2001, Freiry et al. 2020). Community composition and environmental variables were compared among hydrological periods and streams using Permutational Multivariate Analysis of Variance (PERMANOVA), based on a Bray–Curtis dissimilarity matrix (Anderson and Walsh 2013, Freiry et al. 2020). The model included the fixed factors Season, Stream and their interaction. The significance of the terms was assessed using 999 random permutations and the *pseudo*-F statistic (Anderson, 2001). Homogeneity of multivariate dispersion among groups was assessed based on the distance of each sampling unit from the centroid of its respective group in multivariate space, using the *betadisper* function from the **vegan** package (Anderson et al., 2006).

Differences in assemblage composition across the spatial hierarchy and seasons were explored using the Unweighted Pair Group Method with Averages (UPGMA) (Belbin and McDonald 1993, Ávila et al. 2011), based on Bray-Curtis dissimilarities calculated from the zooplankton density matrix (Ávila et al. 2011). Prior to the analyses, density data were relativized and square-root transformed to reduce the influence of highly abundant taxa and improve the distributional properties of the dataset (Rocha et al. 2012). To focus on spatial variation across sampling sites, habitat density data were pooled for this analysis. To assess how well the UPGMA dendrograms represented the original Bray-Curtis dissimilarities among sampling

sites, the Cophenetic Correlation Coefficient (CC) was calculated (Sokal and Rohlf 1962, Sneath and Sokal 1973, Saraçlı et al. 2013), where a cophenetic matrix was constructed from each seasonal dendrogram using the *cophenetic* function in the **vegan** package in R (Oksanen et al. 2020). The cophenetic distance matrix was then correlated with the corresponding original Bray-Curtis dissimilarity matrix using Pearson’s product-moment correlation coefficient (Sokal and Rohlf 1962, Farris 1969, Silva and Dias 2013). To evaluate the strength of the hierarchical clustering structure obtained from the UPGMA dendrograms, the Agglomerative Coefficient (AC) was calculated for each season using the *coef.hclust* function from the **cluster** package in R (Kaufman and Rousseeuw 1990, Maechler et al. 2026). The AC ranges from 0 to 1, where values closer to 1 indicate a strong, well-defined clustering structure and values near 0 reflect weak or indistinct grouping.

Additive diversity partitioning was used to quantify the contribution of within- and between-unit diversity to total zooplankton diversity. In this framework, alpha diversity (α) represents diversity occurring within sampling units, whereas beta diversity (β) represents the contribution of differences among units at successive levels of the spatial hierarchy (Lande 1996, Crist et al. 2003). The hierarchical structure included three spatial scales. The finest scales consisted of the habitat samples within each site ($n = 3$), and of each site ($n = 6$, for the Barro Branco stream, and $n = 9$, for the Caiana stream). The intermediate scales corresponded to the upper and lower stream sections ($n = 2$), whereas the broadest scale represented differences among the two streams. Accordingly, the total diversity (γ) was decomposed into within-habitat diversity (α_1) and the successive beta components associated with differences among habitats (β_1), among sites within upper and lower stream reaches (β_2), and among streams (β_3). The contribution of each component was expressed as a percentage of total diversity (Crist et al. 2003, Farias et al. 2020). To investigate seasonal variation in the hierarchical structure of diversity, additive partitioning was performed separately for each season and for each stream (Melo and Medeiros 2013, Farias et al. 2020).

The overall diversity was taken to be the sum of the relative contributions of each scale in the hierarchical sampling design. Analysis of significance of the contribution of each hierarchical level was performed through individual-based randomizations (type I null model) (see Freiry et al. 2020, Melo and Medeiros 2013). The randomization process was repeated 999 times to obtain null distributions of the α and β diversity at each hierarchical level, and used to test whether the observed values were higher or lower than expected by chance (Crist et al. 2003). The probabilities obtained from randomization tests were interpreted as P -values, as a low P -value ($\text{Prop}_{exp} > \text{Prop}_{obs} < 0.05$) indicates an observed diversity significantly different than that expected under the null model. On the other hand, a high P -value ($\text{Prop}_{exp} > \text{Prop}_{obs} > 0.95$) indicates an observed diversity not different than that expected by chance (Veech et al. 2002, Gering et al. 2003, Crist et al. 2003, Freiry et al. 2020). The results of the analysis were subject to a Monte Carlo permutation test (999 permutations) for validation. The additive partitioning of the diversity was carried out with the function *adipart* from package **vegan** (Oksanen et al. n.d.). All analyses were conducted in the statistical environment R (R Core Team 2017).

3 Results

Water velocity ranged from 0 to 0.41 m/s, with the highest average recorded at site 14. Waters were, on average, well oxygenated (5.3 to 8.1 mg/L), and cool, with average temperatures between from 25.2 to 32.5 °C, the average being 27.7 °C. Conductivity ranged from 98.8 and 223.0 S/cm, pH was mostly neutral to slightly acidic (from 5.1 to 6.9) and turbidity was low (4.9 to 79.2 NTU). Regarding the morphological variables, elevation ranged from 41.5 to 112.8 m a.s.l., while the stream margins slope reached 50° for some sites. Stream width and depth showed greater variations reflecting the range of habitats along the study streams, from small and narrow streams to wider and deeper manmade reservoirs in the stream channel. Stream site width varied from 1 m across to 19.5 m in larger reservoirs. Overall, depths ranged from approx. 10-20 cm to a maximum of 1.55 m, on average. Substrate composition across sites consisted mainly of mud (reaching 100% at sites 1 and 2, in the Barro Branco stream) and sand being more common at the Caiana Stream (reaching 98.3% cover at site 14) (Table 1). Marginal and underwater habitat structures were diverse with most variables being common at the Barro Branco stream, namely aquatic macrophytes, filamentous and attached algae and littoral grass. Submerged and overhanging vegetation, leaf litter, root masses and wood debris were present at all sites (see Table 1).

```
1 #TABELA DE HABITAT----
2
3 dev.off()
4 rm(list=ls(all=TRUE))
5 cat("\014")
6
7 habitat <- read.xlsx("dados/rebio23-habitat.xlsx",
8                     rowNames = T,
9                     colNames = T,
10                    sheet = "ambiente",
11                    startRow = 2)
12 m_hab_part <- read.csv("res/m_hab_part.csv",
13                      sep = ";", dec = ".",
14                      row.names = 1,
15                      header = TRUE,
16                      na.strings = NA)
17 t_grps <- read.csv("res/t_grps.csv",
18                  sep = ";", dec = ".",
19                  row.names = 1,
20                  header = TRUE,
21                  na.strings = NA)
22
23 m_hab_part$w.Temp_C <- habitat[rownames(m_hab_part), "w.Temp_C"]
24 m_hab_part$w.DO_mg.L <- habitat[rownames(m_hab_part), "w.DO_mg.L"]
```

```

25
26 library(dplyr)
27 library(tidyr)
28 m_trab <- m_hab_part %>%
29   rename_with(~ gsub("_", ".", .))
30 m <- m_trab %>%
31   group_by(PontoN = t_grps$PontoN) %>%
32   summarise(across(where(is.numeric),
33                     list(mean = ~ mean(.x, na.rm = TRUE), #ajustes celulas vazias
34                           min = ~ min(.x, na.rm = TRUE),
35                           max = ~ max(.x, na.rm = TRUE))),
36             .groups = 'drop') %>%
37   pivot_longer(
38     cols = -c(PontoN),
39     names_to = c("Variable", ".value"),
40     names_sep = "_"
41   )
42
43 m <- as.data.frame(m)
44 m_wide <- m %>%
45   mutate(stat_string = ifelse(Variable == c("m.Vel.m.s"),
46                                paste0(round(mean, 3), " (", round(min, 3), "-", round(max, 3)
47                                paste0(round(mean, 1), " (", round(min, 1), "-", round(max, 1)
48   unite("Location", PontoN, sep = "_") %>%
49   select(Variable, Location, stat_string) %>%
50   pivot_wider(names_from = Location, values_from = stat_string)
51
52 m_wide
53 m_wide <- as.data.frame(m_wide)
54 cols <- colnames(m_wide)
55 col_order <- c(
56   "Variable",
57   cols[cols != "Variable"][
58     order(as.numeric(gsub("Ponto", "", cols[cols != "Variable"])))
59   ]
60 )
61 m_wide <- m_wide[, c(col_order)] #sequência das colunas
62 row_order <- c(
63   "w.Temp.C",
64   "w.DO.mg.L",
65   "w.Vel.m.s",
66   "w.Cond.uS.cm",
67   "w.Turb.ntu",

```

```

68 "w.pH",
69 "m.Elev.m",
70 "m.Slope",
71 "m.Depth.avg.cm",
72 "m.Depth.mar.cm",
73 "m.Depth.max.cm",
74 "m.Width.m",
75 "s.Mud",
76 "s.Sand",
77 "s.Gravel.f",
78 "s.Rocks",
79 "h.Macrophytes",
80 "h.Algae",
81 "h.Subm.veg",
82 "h.Overh.veg",
83 "h.Grass",
84 "h.Leaf.l",
85 "h.Root.masses",
86 "h.Debris"
87 )
88 m_wide <- m_wide[match(row_order, m_wide$Variable), ]
89 m_wide
90
91 # Salvado m_wide
92 write.table(m_wide, "res/m_wide_hab_to.txt",
93             sep = ";", dec = ".", #"\t",
94             row.names = TRUE,
95             quote = TRUE,
96             append = FALSE)
97 m_wide_hab_to <- read.table("res/m_wide_hab_to.txt",
98                             sep = ";", dec = ".",
99                             row.names = 1,
100                             header = TRUE,
101                             na.strings = NA)
102
103 # Exportando dados para Excel
104 library(openxlsx)
105 #write.xlsx(m_wide, file = "res/tabela de habitat.xlsx", rowNames = FALSE)
106 wb <- loadWorkbook("res/tabela de habitat.xlsx")
107 writeData(wb, sheet = "Sheet 1", x = m_wide)
108 saveWorkbook(wb, "res/tabela de habitat.xlsx", overwrite = TRUE)
109
110 # Tabela final ajustada no Excel

```

```

111 library(readxl)
112 hab_gtttable <- read_excel(
113   path = "res/tabela de habitat.xlsx",
114   sheet = "tabela_final")
115 #range = "A1:P29"#ajustar para o tamanho da tabela final
116 hab_gtttable
117 library(gt)
118 hab_gtttable <- gt(hab_gtttable)
119 hab_gtttable <- sub_missing(hab_gtttable,
120                             columns = everything(), missing_text = "")
121 #hab_gtttable <- fmt_number(hab_gtttable, columns = "F.O.(%)", decimals = 1)
122 hab_gtttable
123 gtsave(hab_gtttable, "res/gt-hab_gtttable.xlsx.html")
124 saveRDS(hab_gtttable, "res/gt-hab_gtttable.xlsx.rds")

1 #TABELA DE HABITAT----
2 ##SEGUE DO SCRIPT ANTERIOR
3
4 # Escolher sumário de uma variavel
5 m
6 var <- "w.Vel.m.s"
7 m[m$Variable == var, "mean"] #cada valor de var
8 summary(m[m$Variable == var, "mean"]) #sumário dos valores de var
9
10 # Escolher sumário de um grupo de variáveis do df m
11 vars <- unique(grep("^h\\.\\.", m$Variable, value = TRUE))
12 summaries <- list() #cria uma lista vazia para guardar os sumários
13 # Loop para cada variável do grupo e guarda em summaries
14 for (var in vars) {
15   summaries[[var]] <- summary(m[m$Variable == var, "mean"])
16 }
17 #var is a temporary variable used in the for loop to iterate through
18 #each variable name that starts with "h."
19
20 summaries
21 summary_table <- do.call(rbind, lapply(summaries, as.data.frame.list))
22 colnames(summary_table) <- c("Min", "Q1", "Median", "Mean", "Q3", "Max")
23 #sink(file = "res/summary_w.txt", split = TRUE)
24 round(summary_table, 2)
25 #round(summary_table[order(summary_table$Mean, decreasing = FALSE), ], 2)
26 #sink()
27
28 #Sumario para as médias

```

```

29
30 dev.off()
31 rm(list=ls(all=TRUE))
32 cat("\014")
33
34 m_hab_part <- read.csv("res/m_hab_part.csv",
35                       sep = ";", dec = ".",
36                       row.names = 1,
37                       header = TRUE,
38                       na.strings = NA)
39 t_grps <- read.csv("res/t_grps.csv",
40                   sep = ";", dec = ".",
41                   row.names = 1,
42                   header = TRUE,
43                   na.strings = NA)
44
45 library(dplyr)
46 library(tidyr)
47
48 m_trab <- m_hab_part %>%
49   rename_with(~ gsub("_", ".", .))
50
51 # Compute averages per group
52 m_s <- m_trab %>%
53   group_by(PontoN = t_grps$PontoN) %>%
54   summarise(across(where(is.numeric), ~ mean(.x, na.rm = TRUE)),
55             .groups = 'drop') %>%
56   pivot_longer(
57     cols = -PontoN,
58     names_to = "Variable",
59     values_to = "mean"
60   )
61
62 # Format rounding and reshape to wide format
63 m_wide_s <- m_s %>%
64   mutate(
65     stat_string = ifelse(
66       Variable == "m.Vel.m.s",
67       as.character(round(mean, 3)),
68       as.character(round(mean, 1))
69     )
70   ) %>%
71   unite("Location", PontoN, sep = "_") %>%

```

```

72   select(Variable, Location, stat_string) %>%
73   pivot_wider(names_from = Location, values_from = stat_string)
74
75 m_wide_s <- as.data.frame(m_wide_s)
76
77 # Order columns numerically by Ponto number
78 cols <- colnames(m_wide_s)
79 col_order <- c(
80   "Variable",
81   cols[cols != "Variable"][
82     order(as.numeric(gsub("Ponto", "", cols[cols != "Variable"])))]
83 ]
84 )
85 m_wide_s <- m_wide_s[, col_order]
86
87 # Order rows according to pre-defined variable list
88 row_order <- c(
89   "w.Cond.uS.cm",
90   "w.Turb.ntu",
91   "w.pH",
92   "m.Elev.m",
93   "m.Vel.m.s",
94   "m.Slope",
95   "m.Depth.avg.cm",
96   "m.Depth.mar.cm",
97   "s.Mud",
98   "s.Sand",
99   "s.Gravel.f",
100  "s.Rocks",
101  "h.Macrophytes",
102  "h.Algae",
103  "h.Subm.veg",
104  "h.Overh.veg",
105  "h.Grass",
106  "h.Leaf.l",
107  "h.Root.masses",
108  "h.Debris"
109 )
110
111 m_wide_s <- m_wide_s[match(row_order, m_wide_s$Variable), ]
112
113 # View final table
114 m_wide_s

```

Principal Component Analysis (PCA) was used to summarize environmental variation among sampling sites and to identify variables contributing to the differences between streams (Figure 2) and seasons (Figure 3). The first two principal components explained 38.6% of the total variance, with the first axis accounting for 23.2%. In relation to the environmental variables of water quality, the Barro Branco stream was more associated with higher elevations, the predominance of muddy substrate and a narrower array of underwater structures, mostly algae, macrophytes, leaf litter and woody debris. The Caiana stream showed a wider range of variables affecting its waters, being associated with higher pH, turbidity and depths, as well as a predominance of sandy substrates and, to a lesser degree, root masses. Water velocity was also important for this stream. PERMANOVA showed significant differences in the environmental variables between the two study streams ($pseudo-F_{1,86} = 8.3, R^2 = 0.082, p = 0.001$). Even though dry and wet seasons did not show clear segregation in the PCA plot, PERMANOVA indicated significance of differences ($pseudo-F_{1,86} = 4.2, R^2 = 0.042, p = 0.006$). A significant Season $\times$ Stream interaction was also detected ($pseudo-F_{1,86} = 2.8, R^2 = 0.028, p = 0.032$), indicating that the effect of hydrological period on the environmental variables differed between Barro Branco and Caiana. The complete model explained 15.1% of the variation, whereas 84.9% remained unexplained by the factors included in the model.

```

1  #PCA fviz package----
2  #browseURL("https://www.sthda.com/english/articles/31-principal-component-methods-in-r-pract
3
4  #ORGANIZANDO DADOS----
5
6  dev.off()
7  rm(list=ls(all=TRUE))
8  cat("\014")
9
10 t_grps <- read.csv("res/t_grps.csv",
11                   sep = ";", dec = ".",
12                   row.names = 1,
13                   header = TRUE,
14                   na.strings = NA)
15 m_hab_trns <- read.csv("res/m_hab_trns.csv",
16                      sep = ";", dec = ".",
17                      row.names = 1,
18                      header = TRUE,
19                      na.strings = NA)
20
21 #PCA----
22
23 library("FactoMineR")
24 library("factoextra")
25

```

```

26  pca <- PCA(m_hab_trns, scale.unit = TRUE, ncp = 5, graph = TRUE)
27  print(pca)
28
29  eig.val <- get_eigenvalue(pca)
30  eig.val
31
32  #sink(file = "res/cor_matrix.txt", append = FALSE)
33  pca$var$cor
34  #sink()
35
36  fviz_eig(pca, addlabels = TRUE, ylim = c(0,30))
37
38  var <- get_pca_var(pca)
39  var
40
41  fviz_pca_var(pca, col.var = "black")
42
43  library("corrplot")
44  corrplot(var$cos2, is.corr=FALSE)
45
46  fviz_cos2(pca, choice = "var", axes = 1:2)
47
48  fviz_pca_var(pca, col.var = "cos2",
49              gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
50              repel = TRUE # Avoid text overlapping
51  )
52
53  fviz_pca_var(pca, alpha.var = "cos2")
54
55  corrplot(var$contrib, is.corr=FALSE)
56
57  fviz_contrib(pca, choice = "var", axes = 1, top = 10)
58  fviz_contrib(pca, choice = "var", axes = 2, top = 10)
59  fviz_contrib(pca, choice = "var", axes = 1:2, top = 10)
60
61  fviz_pca_var(pca, col.var = "contrib",
62              gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07")
63  )
64  fviz_pca_var(pca, alpha.var = "contrib")
65
66  ##GROUPING BY KMEANS----
67
68  set.seed(123)

```

```

69 res.km <- kmeans(var$coord, centers = 3, nstart = 25)
70 grp <- as.factor(res.km$cluster)
71 # Color variables by groups
72 fviz_pca_var(pca, col.var = grp,
73               palette = c("#0073C2FF", "#EFC000FF", "#868686FF"),
74               legend.title = "Cluster")
75
76 ind <- get_pca_ind(pca)
77 ind
78 ind$contrib
79
80 ##BIPLOTS----
81
82 fviz_pca_ind(pca)
83
84 fviz_pca_ind(pca, col.ind = "cos2",
85               gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
86               repel = TRUE #avoid text overlapping (slow if many points)
87 )
88
89 fviz_pca_ind(pca, pointsize = "cos2",
90               pointshape = 21, fill = "#E7B800",
91               repel = TRUE #avoid text overlapping (slow if many points)
92 )
93
94 fviz_pca_ind(pca, col.ind = "cos2", pointsize = "cos2",
95               gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
96               repel = TRUE #avoid text overlapping (slow if many points)
97 )
98
99 fviz_cos2(pca, choice = "ind")
100
101 fviz_contrib(pca, choice = "ind", axes = 1:2)
102
103 # Create a random continuous variable of length 23,
104 # Same length as the number of active individuals in the PCA
105 t_grps <- t_grps[rownames(pca$ind$coord), ]
106 colnames(t_grps)
107 set.seed(123)
108 # Escolhendo variável categorica
109 my.cont.var <- t_grps$Periodo
110 my.cont.var[my.cont.var == "seco"] <- "dry"
111 my.cont.var[my.cont.var == "chuvoso"] <- "wet"

```

```

112 # Color individuals by the continuous variable
113 fviz_pca_ind(
114   pca,
115   habillage = my.cont.var,
116   palette = c("blue", "yellow", "red", "pink"),
117   repel = FALSE
118 )
119
120 fviz_pca_ind(pca,
121               geom.ind = "point", #show points only (nbut not "text")
122               col.ind = as.factor(my.cont.var), #color by groups
123               palette = "grey",
124               addEllipses = TRUE, #concentration ellipses
125               legend.title = "Groups"
126 )
127
128 length(unique(my.cont.var))
129 fviz_pca_ind(pca,
130               geom.ind = "point",
131               col.ind = my.cont.var,
132               palette = "grey",
133               addEllipses = TRUE, ellipse.type = "confidence",
134               mean.point = TRUE,
135               legend.title = "Groups")
136
137 fviz_pca_biplot(pca, repel = TRUE,
138                 col.var = "#2E9FDF", #variables color
139                 col.ind = "#696969" #individuals color
140 )
141
142 # PCA com ellipse colorida
143 fviz_pca_biplot(pca,
144                 col.ind = my.cont.var, palette = "jco",
145                 addEllipses = TRUE, ellipse.type = "t", #t-distribution confidence ellipses"
146                 ellipse.level = 0.95,
147                 shape.ind = my.cont.var,
148                 label = "var",
149                 col.var = "black", repel = TRUE,
150                 legend.title = "Stream") +
151 ggtitle(NULL) +
152 scale_color_manual(values = c(
153   "#1B9E77",
154   "#D95F02",

```

```

155     "#7570B3",
156     "#E7298A"
157   )) +
158   scale_shape_manual(values = c(
159     16, # circle
160     17, # triangle
161     15, # square
162     18 # diamond
163   ))
164
165   ### FILTRANDO AS VARIÁVEIS
166   var_coord <- pca$var$coord[,1:2]
167   keep_vars <- apply(abs(var_coord) > 0.2,1,any)
168   vars_to_keep <- rownames(var_coord)[keep_vars]
169   pca_filt <- pca
170   pca_filt$var$coord <- pca$var$coord[vars_to_keep,,drop=FALSE]
171   pca_filt$var$cos2 <- pca$var$cos2[vars_to_keep,,drop=FALSE]
172   pca_filt$var$contrib <- pca$var$contrib[vars_to_keep,,drop=FALSE]
173
174   ### AJUSTE MANUAL DAS COORDENADAS DOS VETORES
175
176   # Cria uma tabela SEPARADA apenas para as posições dos rótulos
177   label_coords <- as.data.frame(pca_filt$var$coord)
178   # Multiplica APENAS a posição do texto (as setas continuam intactas no pca_filt)
179   label_coords <- label_coords * 2.0
180   label_coords <- label_coords * 2.0
181   # Se quiser ajustar pontos específicos manualmente
182   #fix(label_coords) #nomes das colunas
183   #write.table(label_coords,"res/coords_lab.txt")
184   label_coords <- read.table("res/coords_lab.txt",
185                             header=TRUE,
186                             row.names=1)
187
188   # GRÁFICO FINAL
189
190   #png("res/fig-hab_PCA_periodes.png", width = 9, height = 7.5, units = "in", res = 300)
191   fviz_pca_biplot(pca_filt,
192     col.ind = my.cont.var, palette = "jco",
193     addEllipses = TRUE, ellipse.type = "t",
194     ellipse.level = 0.95,
195     shape.ind = my.cont.var,
196     label = "none",
197     col.var = "black",

```

```

198   legend.title = "Season"
199 ) +
200   # Adiciona os rótulos nas posições customizadas da tabela label_coords
201   geom_text(
202     data = label_coords,
203     aes(x = Dim.1, y = Dim.2, label = rownames(label_coords)),
204     color = "black",
205     size = 12) +
206   ggtitle(NULL) +
207   scale_color_manual(values = c("#1B9E77", "#D95F02", "#7570B3", "#E7298A")) +
208   scale_fill_manual(values = c("darkcyan", "orange", "#7570B3", "#E7298A")) +
209   scale_shape_manual(values = c(
210     16, # circle
211     17, # triangle
212     15, # square
213     18 # diamond
214   )) +
215   theme(
216     axis.title = element_text(size = 34),
217     axis.text = element_text(size = 32),
218     legend.title = element_text(size = 40),
219     legend.text = element_text(size = 40)
220   )
221   #dev.off()

1  #ORGANIZANDO DADOS----
2
3  dev.off()
4  rm(list=ls(all=TRUE))
5  cat("\014")
6
7  t_grps <- read.csv("res/t_grps.csv",
8                    sep = ";", dec = ".",
9                    row.names = 1,
10                   header = TRUE,
11                   na.strings = NA)
12  m_hab_trns <- read.csv("res/m_hab_trns.csv",
13                        sep = ";", dec = ".",
14                        row.names = 1,
15                        header = TRUE,
16                        na.strings = NA)
17
18  colnames(t_grps)

```

```

19 m_trab <- m_hab_trns
20 library(vegan)
21
22 #MRPP----
23
24 #TRABALHANDO COM MULTIPLAS VARIAVEIS----
25
26 # Run MRPP for multiple variables
27 fatores <- c("Periodo", "Rio", "Ref_site")
28 set.seed(123)
29 mrpp_results <- lapply(fatores, function(v) {
30   grp <- as.factor(t_grps[[v]])
31   mrpp(
32     dat = m_trab,
33     grouping = grp,
34     distance = "bray"
35   )
36 })
37 names(mrpp_results) <- fatores
38 mrpp_results
39 # Print one result
40 mrpp_results$Ref_site
41 # Loop through summaries
42 sink(file = "res/mrpp-env.txt", split = TRUE)#, append = TRUE)
43 for (v in names(mrpp_results)) {
44   cat("\n=====\\n")
45   cat("MRPP for:", v, "\\n")
46   print(mrpp_results[[v]])
47 }
48 sink()
49 # Extract key statistics into a table
50 mrpp_summary <- data.frame(
51   Variable = fatores,
52   Delta = sapply(mrpp_results, function(x) x$delta),
53   Expected_Delta = sapply(mrpp_results, function(x) x$E.delta),
54   A = sapply(mrpp_results, function(x) x$A),
55   P_value = sapply(mrpp_results, function(x) x$Pvalue)
56 )
57 sink("res/mrpp-env.txt", append = TRUE)
58 mrpp_summary
59 sink()
60
61 ##TRABALHANDO COM UMA UNICA VARIABEL----

```

```

62
63 # Run MRPP for a single variable
64 grp <- as.factor(t_grps$Ref_site) #UA, mes, estacao, area, ambiente
65 set.seed(123)
66 mrpp <- mrpp(dat = m_trab, grouping = grp, distance = "bray")
67 mrpp
68 summary(mrpp)
69
70 # Teste de hipótese para uma unica variavel
71 str(mrpp)
72 p_value <- mrpp$Pvalue
73 print(sprintf("p-value: %.10f", p_value))
74 # Conditional statement to check the p-value and print the appropriate message
75 if (p_value < 0.05) {
76   print(sprintf("Existe diferença significativa porque o valor de p é %.10f, sendo menor que
77 } else {
78   print(sprintf("Não existe diferença significativa porque o valor de p é %.10f, sendo maior
79 }
80
81 ##TESTES PAREADOS----
82
83 # Todos os pares de níveis do fator
84 pairs <- combn(levels(as.factor(grp)), 2, simplify = FALSE)
85 pairs
86 # Aplicar mrpp para cada par
87 res_list <- lapply(pairs, function(p) {
88   sel <- grp %in% p
89   mrpp(dat = m_trab[sel, ], grouping = droplevels(grp[sel]), distance = "bray")
90 })
91 # Dar nomes aos resultados
92 names(res_list) <- sapply(pairs, function(p) paste(p, collapse = "_vs_"))
93 # visualizar
94 res_list
95 # Resumo
96 summary_tab <- data.frame(
97   Comparison = names(res_list),
98   A = sapply(res_list, function(x) x$A),
99   Delta = sapply(res_list, function(x) x$delta),
100   Pvalue = sapply(res_list, function(x) x$Pvalue)
101 )
102 sink("res/mrpp-env.txt", append = TRUE)
103 summary_tab
104 sink()

```

```

105
106 ###MATRIX-STYLE TABLE----
107 library(dplyr)
108 library(tidyr)
109 library(knitr)
110 library(kableExtra)
111 # Extract unique group names from the comparisons
112 groups1 <- sort(unique(unlist(strsplit(summary_tab$Comparison, "_vs_"))))
113 # Initialize empty matrix
114 pval_matrix <- matrix(NA, nrow = length(groups1), ncol = length(groups1),
115                       dimnames = list(groups1, groups1))
116 # Fill matrix with Pvalues
117 for (i in 1:nrow(summary_tab)) {
118   comp <- unlist(strsplit(summary_tab$Comparison[i], "_vs_"))
119   g1 <- comp[1]
120   g2 <- comp[2]
121   p <- summary_tab$Pvalue[i]
122
123   pval_matrix[g1, g2] <- p
124   pval_matrix[g2, g1] <- p #make symmetric
125 }
126
127 # Format p-values: red if <0.05
128 pval_colored <- ifelse(is.na(pval_matrix), "",
129                       ifelse(pval_matrix < 0.05,
130                             paste0("<span style='color:red;'>", sprintf("%.3f", pval_matrix[i, j])),
131                             sprintf("%.3f", pval_matrix[i, j])))
132
133 # Convert to data.frame for kable
134 pval_df <- as.data.frame(pval_colored)
135 pval_df <- tibble::rownames_to_column(pval_df, "Group")
136 # Render HTML table
137 matrixstyle <- pval_df %>%
138   kable("html", escape = FALSE) %>%
139   kable_styling(full_width = FALSE)
140 matrixstyle
141 save_kable(matrixstyle, "res/kb-mrpp-env_matrixstyle.html")
142
143 ##GRÁFICO DA MRPP----
144
145 png("res/fig-mrpp-env.png", width = 15, height = 10, units = "in", res = 300)
146 def.par <- par(no.readonly = TRUE)
147 layout(matrix(1:2,nr=1))

```

```

148 plot(ord <- metaMDS(m_trab, distance="bray"), type="text", display="sites")
149 with(t_grps, ordihull(ord, grp))
150 with(mrpp, {
151   fig.dist <- hist(boot.deltas, xlim=range(c(delta,boot.deltas)),
152                   main="Test of Differences Among Groups")
153   abline(v=delta);
154   text(delta, 2*mean(fig.dist$counts), adj = -0.5,
155         expression(bold(delta)), cex=1.5 ) }
156 )
157 par(def.par)
158 dev.off()
159
160 ###AVALIANDO AS DISTÂNCIAS MÉDIAS----
161 md <- with(t_grps, meandist(vegdist(m_trab), grp))
162 md
163 summary(md)
164 par(mfrow=c(1,2))
165 plot(md)
166 plot(md, kind="histogram")
167 par(mfrow=c(1,1))
168
169 #PerMANOVA----
170
171 #browseURL("https://riffomonas.org/code_club/2021-03-18-vegan")
172 #browseURL("https://riffomonas.org/code_club/2021-03-22-reproducibility")
173
174 library(tidyverse)
175 library(ggtext)
176 library(ggplot2)
177 library(vegan)
178
179 ## TESTES PARA VARIÁVEIS COM MAIS DE DOIS FATORES
180
181 adonis2(m_trab ~ Ref_site, data = t_grps, method = "bray")
182 set.seed(123)
183 perm <- 999
184 all_test <- adonis2(m_trab ~ Ref_site, data = t_grps, method = "bray", permutations = perm)
185 all_test
186
187 ## TESTES PARA VARIÁVEIS COM DOIS FATORES
188
189 sink(file = "res/perma-env.txt", split = TRUE)#, append = TRUE)
190 set.seed(123)

```

```

191 perm <- 999
192 adonis2(m_trab ~ Período, data = t_grps, method = "bray", permutations = perm)
193 adonis2(m_trab ~ Rio, data = t_grps, method = "bray", permutations = perm)
194 adonis2(m_trab ~ Período + Rio, data = t_grps, #+ no lugar de * tira os resíduos
195         method = "bray", permutations = perm)
196
197 ## AVALIANDO INTERAÇÕES-----
198
199 adonis2(m_trab ~ Período*Rio, data = t_grps, method = "bray", permutations = perm,
200         by = "terms") #ou by=NULL remove as interações
201 adonis2(m_trab ~ Período*Rio, data = t_grps, method = "bray", permutations = perm,
202         by = "margin") #ou by=NULL remove as interações
203 adonis2(m_trab ~ Período*Rio, data = t_grps, method = "bray", permutations = perm,
204         by = NULL) #ou by=NULL remove as interações
205 sink()
206
207 #BETADISPER
208
209 dist_bray <- vegdist(m_trab, method = "bray")
210 t_grps$Período_Rio <- interaction(
211     t_grps$Período,
212     t_grps$Rio,
213     sep = "_")
214 bd <- betadisper(dist_bray, t_grps$Período_Rio)
215 permutest(bd, permutations = 999)
216
217 # PARA CADA FATOR
218
219 bd_período <- betadisper(dist_bray, t_grps$Período)
220 bd_rio <- betadisper(dist_bray, t_grps$Rio)
221 permutest(bd_período, permutations = 999)
222 permutest(bd_rio, permutations = 999)
223
224 ##### ARRUMAR ISSO #####
225
226 all_p <- all_test$`Pr(>F)`[1]
227 all_p
228
229 count(t_grps, Período)
230 colnames(t_grps)
231
232 # Pairwise comparisons
233 #browseURL("https://doi.org/10.1007/s10641-018-0751-1")

```

```

234 #Lanés, L.E.K., Reichard, M., de Moura, R.G. et al. 2018.
235 #Environmental predictors for annual fish assemblages in subtropical
236 #grasslands of South America: the role of landscape and habitat
237 #characteristics. Environ Biol Fish 101, 963-977.
238
239 library(dplyr)
240 #library(combninat) # or base R combn()
241
242 set.seed(123)
243
244 # Get unique levels of UA
245 groups <- unique(t_grps$Ref_site)
246 groups
247 # All pairwise combinations
248 pairs <- combn(groups, 2, simplify = FALSE)
249 pairs
250 # Store results
251 pairwise_results <- lapply(pairs, function(g) {
252   # Subset data
253   sub_data <- t_grps %>% filter(Ref_site %in% g)
254   sub_dist <- m_trab[rownames(sub_data), ]
255
256   # Run adonis2
257   test <- adonis2(sub_dist ~ Ref_site, data = sub_data,
258                   method = "bray", permutations = perm)
259
260   tibble(
261     group1 = g[1],
262     group2 = g[2],
263     F_value = test$F[1],
264     R2 = test$R2[1],
265     p_value = test$`Pr(>F)`[1]
266   )
267 })
268 # Bind all results
269 pairwise_results <- bind_rows(pairwise_results)
270 pairwise_results
271 pairwise_p <- numeric()
272 pairwise_p <- pairwise_results$p_value
273 pairwise_p
274 adj_p_values <- p.adjust(pairwise_p, method="bonferroni") #c("holm", "hochberg", "hommel", "
275 adj_p_values
276 pairwise_results$adj_p_values <- adj_p_values

```

```

277 pairwise_results
278 # Get unique group names
279 groups2 <- sort(unique(c(pairwise_results$group1, pairwise_results$group2)))
280 # Initialize empty matrix
281 pval_matrix <- matrix(NA, nrow = length(groups2), ncol = length(groups2),
282                       dimnames = list(groups2, groups2))
283 # Fill in p-values
284 for (i in 1:nrow(pairwise_results)) {
285   g1 <- pairwise_results$group1[i]
286   g2 <- pairwise_results$group2[i]
287   p <- pairwise_results$p_value[i]
288
289   pval_matrix[g1, g2] <- p
290   pval_matrix[g2, g1] <- p #make symmetric
291 }
292 pval_matrix
293
294 library(knitr)
295 library(kableExtra)
296
297 # Make a copy of the matrix
298 pval_colored <- pval_matrix
299 # Replace numeric values with formatted HTML strings
300 pval_colored[] <- ifelse(is.na(pval_matrix), "",
301                          ifelse(pval_matrix < 0.05,
302                                paste0("<span style='color:red;'>", sprintf("%.3f", pval_matr
303                                sprintf("%.3f", pval_matrix)))
304 # Convert to data frame with rownames
305 pval_df <- as.data.frame(pval_colored)
306 pval_df <- tibble::rownames_to_column(pval_df, "Group")
307 # Render table
308 pval_df %>%
309   kable("html", escape = FALSE) %>%
310   kable_styling(full_width = FALSE)
311 save_kable(matrixstyle, "res/kb-perma_matrixstyle.html")
312
313 ## AVALIANDO INTERAÇÕES-----
314
315 adonis2(m_trab ~ Período, data = t_grps, method = "bray")
316 adonis2(m_trab ~ Período*Rio, data = t_grps,
317         method = "bray") #+ no lugar de * tira os residuos
318 adonis2(m_trab ~ Período*Rio, data = t_grps,
319         method = "bray", by = "terms", strata = NULL)

```

Among the averaged nutrient concentrations (Table 2), ammonia ranged from 49.2 to 243.7 µg/L. The Caiana stream tended to have greater values, but overall no clear trend can be observed when comparing both streams. Nitrite and nitrate, on the other hand, show clearly a greater concentrations at the Caiana stream, nitrate varying from 166.4 to 2003.0 µg/L, compared with the maximum of 18.8 µg/L at the Barro Branco stream. Nitrite was too low to be detectable at the Barro Branco stream. The same pattern was observed for orthophosphate concentrations, being barely detectable at Barro Branco stream, and varying from 4.0 to 32.0 µg/L at the Caiana stream. Total phosphorus, however, tended to have higher concentrations at Barro Branco stream, reaching 50.1 µg/L at Site 3, but with highly variable values (from non detectable to 38.7 µg/L, as the second highest value).

```

1  #TABELA DE NUTRIENTES----
2
3  ##ORGANIZANDO DADOS----
4
5  dev.off()
6  rm(list=ls(all=TRUE))
7  cat("\014")
8
9  t_nutri <- read.csv("res/t_nutri.csv",
10                    sep = ";", dec = ".",
11                    row.names = 1,
12                    header = TRUE,
13                    na.strings = NA)
14
15  t_grps <- read.csv("res/t_grps.csv",
16                    sep = ";", dec = ".",
17                    row.names = 1,
18                    header = TRUE,
19                    na.strings = NA)
20
21  nutri <- t_nutri
22
23  library(dplyr)
24  library(tidyr)
25  t_grps <- t_grps[seq_len(nrow(nutri)), , drop = FALSE]
26  m_trab <- nutri %>%
27    rename_with(~ gsub("_", ".", .))
28  m <- m_trab %>%
29    group_by(PontoN = t_grps$PontoN) %>%
30    summarise(across(where(is.numeric),
31                      list(mean = mean, min = min, max = max)),
32              .groups = 'drop') %>%

```

```

33   pivot_longer(
34     cols = -c(PontoN),
35     names_to = c("Variable", ".value"),
36     names_sep = "_"
37   )
38
39   m <- as.data.frame(m)
40   colnames(m_trab)
41   m_wide <- m %>%
42     mutate(stat_string = ifelse(Variable == c("Ammonia.NH3.ug.L"),
43                                paste0(round(mean, 1), " (", round(min, 1), "-", round(max, 1)
44                                paste0(round(mean, 1), " (", round(min, 1), "-", round(max, 1)
45     unite("Location", PontoN, sep = "_") %>%
46     select(Variable, Location, stat_string) %>%
47     pivot_wider(names_from = Location, values_from = stat_string)
48
49   m_wide
50   m_wide <- as.data.frame(m_wide)
51   cols <- colnames(m_wide)
52   col_order <- c(
53     "Variable",
54     cols[cols != "Variable"] [
55       order(as.numeric(gsub("Ponto", "", cols[cols != "Variable"])))
56     ]
57   )
58   m_wide <- m_wide[, c(col_order)] #sequência das colunas
59   m_wide
60
61   # Salvado m_wide
62   write.table(m_wide, "res/m_wide_nutri.txt",
63     sep = ";", dec = ".", #"\\t",
64     row.names = TRUE,
65     quote = TRUE,
66     append = FALSE)
67   m_wide_nutri <- read.table("res/m_wide_nutri.txt",
68     sep = ";", dec = ".",
69     row.names = 1,
70     header = TRUE,
71     na.strings = NA)
72
73   # Exportando dados para Excel
74   library(openxlsx)
75   #write.xlsx(m_wide, file = "res/tabela de nutrientes.xlsx", rowNames = FALSE)

```

```

76 wb <- loadWorkbook("res/tabela de nutrientes.xlsx")
77 writeData(wb, sheet = "Sheet 1", x = m_wide_nutri)
78 saveWorkbook(wb, "res/tabela de nutrientes.xlsx", overwrite = TRUE)
79
80 # Tabela final ajustada no Excel
81 library(readxl)
82 nutri_gtttable <- read_excel(
83   path = "res/tabela de nutrientes.xlsx",
84   sheet = "tabela_final",
85   range = "A1:P7") #ajustar para o tamanho da tabela final
86 nutri_gtttable
87 library(gt)
88 nutri_gtttable <- gt(nutri_gtttable)
89 nutri_gtttable <- sub_missing(nutri_gtttable,
90                               columns = everything(), missing_text = "")
91 nutri_gtttable
92 gtsave(nutri_gtttable, "res/gt-nutri_gtttable_xlsx.html")
93 saveRDS(nutri_gtttable, "res/gt-nutri_gtttable_xlsx.rds")

```

A total of 56 taxa distributed among Rotifera (14 families), Cladocera (5 families) and Copepoda (2 families) were identified during the study period. Rotifera was the most species-rich group, represented by 39 species (plus Bdelloidea), followed by Cladocera (12 species) and Copepoda (4 species, in addition to the juvenile stages nauplii and copepodite). Across sampling periods, Lecanidae exhibited the highest richness among all recorded families of Rotifera with 15 species, followed by Lepadellidae, Synchaetidae (both with 4 species), Brachionidae and Mytilinidae (both with 3 species). Bdelloidea was the most abundant taxa (6194 individuals) followed by *Lecane curvicornis* with 626 individuals. Among the cladocerans, Chydoridae was the richest family with 5 species, followed by Macrothricidae (3 species). *Alonella dadayi* was the most abundant among the cladocerans with 1051 individuals. Juvenile stages (nauplii and copepodite) were also abundant with 2433 and 4061 individuals, respectively. Copepods were represented by the orders Cyclopoida and Calanoida, *Paracyclops* sp. *Mesocyclops* sp. being the most abundant (141 and 231 individuals, respectively) (Table 3).

```

1  #ZOOEBIO_part - ESTR. DA COMUNIDADE-----
2  #####.....-----
3
4  ##ORGANIZANDO DADOS-----
5
6  dev.off() #apaga os graficos, se houver algum
7  rm(list=ls(all=TRUE)) #limpa a memória
8  cat("\014") #limpa o console
9
10 t_grps <- read.csv("res/t_grps.csv",

```

```

11         sep = ";", dec = ".",
12         row.names = 1,
13         header = TRUE,
14         na.strings = NA)
15 m_dens <- read.csv("res/m_dens.csv",
16                 sep = ";", dec = ".",
17                 row.names = 1,
18                 header = TRUE,
19                 na.strings = NA)
20 m_hab_part <- read.csv("res/m_hab_part.csv",
21                      sep = ";", dec = ".",
22                      row.names = 1,
23                      header = TRUE,
24                      na.strings = NA)
25 m_cont <- read.csv("res/m_cont.csv",
26                  sep = ";", dec = ".",
27                  row.names = 1,
28                  header = TRUE,
29                  na.strings = NA)
30
31 library(tibble)
32 library(tidyverse)
33 library(forcats)
34 library(openxlsx)
35 library(Rcpp)
36
37 m_dens
38 str(m_dens)
39 m_trab <- m_dens
40
41 m_trab <- (m_dens) #SITES
42 m_trab <- t(m_dens) #SPP
43
44 ##TAMANHO DA MATRIZ----
45
46 range(m_trab) #menor e maior valores
47 length(m_trab) #no. de colunas
48 ncol(m_trab) #no. de N colunas
49 nrow(m_trab) #no. de M linhas
50 sum(lengths(m_trab)) #soma os nos. de colunas
51 length(as.matrix(m_trab)) #tamanho da matriz m x n
52 sum(m_trab == 0) #número de observações igual a zero
53 sum(m_trab > 0) #número de observações maiores que zero

```

```

54  #calculando a proporção de zeros na matriz
55  zeros <- (sum(m_trab == 0)/length(as.matrix(m_trab)))*100
56  zeros
57
58  ##ESTRUTURA DA COMUNIDADE----
59
60  Sum <- rowSums(m_trab)
61  #ou
62  Sum <- apply(m_trab,1,sum)
63  Sum
64  # Abundância relativa (%)
65  RA <- (Sum / sum(Sum)) * 100 # percentage
66  # Media
67  Mean <- rowMeans(m_trab)
68  Mean
69  # Ou
70  Mean <- apply(m_trab,1,mean)
71  Mean
72  # Desvio padrão
73  DP <- apply(m_trab,1,sd)
74  DP
75  # Máximo
76  Max <- apply(m_trab,1,max)
77  Max
78  # Mínimo
79  Min <- apply(m_trab,1,min)
80  Min
81  # Mínimo não-zero
82  MinZ <- apply(m_trab, 1, function(row) {
83    non_zero_values <- row[row > 0] # Filter out zero values
84    if (length(non_zero_values) == 0) {
85      return(0) # If all values are zero, return 0
86    } else {
87      return(min(non_zero_values)) # Return the minimum of non-zero values
88    }
89  })
90  MinZ
91
92  m_pa <- m_trab
93  m_pa[m_pa != 0] <- 1
94  rowSums(m_pa)
95  library(vegan)
96  bin <- decostand(m_trab,"pa")

```

```

97  bin[1:10, 1:10]
98  S <- apply(bin,1,sum)
99  S
100  #OU
101  Riqueza <- specnumber(m_trab)
102  Riqueza
103  Riqueza_total <- specnumber(colSums(m_trab))
104  Riqueza_total
105
106  FO <- rowSums(m_trab > 0) / ncol(m_trab) * 100
107  FO
108
109  H <- diversity(m_trab, index = "shannon")
110  H
111  D <- diversity(m_trab, "simpson")
112  D
113  D[is.na(D)] <- 0 #substitui NA ou NaN por 0
114  D
115  E <- H/log(specnumber(m_trab))
116  E
117  E[is.na(E)] <- 0 #substitui NA ou NaN por 0
118  E
119
120  ###DESCRITORES-----
121  Descritores1 <- cbind(Sum, RA, Mean, DP, Max, Min, MinZ, FO, S, E, H, D)
122  Descritores1 <- as.data.frame(Descritores1)
123  Descritores1
124  Descritores1 <- Descritores1[order(-Descritores1$Mean), ] #ordena do maior para o menor "-"
125  #Descritores1 <- Descritores1 %>% rownames_to_column(var="Espécies") #da nome a primeira col
126  SomaTotalD <- apply(Descritores1,2,sum)
127  SomaTotalD
128  MediaTotalD <- apply(Descritores1,2,mean)
129  MediaTotalD
130  DPTotalD <- apply(Descritores1,2,sd)
131  DPTotalD
132  Descritores2 <- cbind(SomaTotalD, MediaTotalD, DPTotalD)
133  Descritores2 <- as.data.frame(Descritores2)
134  Descritores2 <- t(Descritores2)
135  Descritores2
136  DescritoresFinal <- rbind(Descritores1, Descritores2)
137  DescritoresFinal
138  DescritoresFinal <- round (DescritoresFinal, 2)
139  DescritoresFinal

```

```

140
141 # Salvando DescritoresFinal
142 write.table(data.frame("SPP"=rownames(DescritoresFinal),
143                       DescritoresFinal),
144             "res/DescritoresSPP.txt",
145             row.names=FALSE,
146             sep="\t")
147
148 ###TABELA FINAL DESCRITORES-----
149 library(gt)
150 df <- DescritoresFinal
151 ncol(df); nrow(df) #no. de N colunas x M linhas
152 df <- cbind(SPP = rownames(df), df)
153 gt <- gt(df, rowname_col = "SPP",
154          caption = "Descritores da diversidade por espécie (colunas). Sum, soma; RA, abundância re
155 gt
156 gtsave(gt, "res/DescritoresSPP.html")
157 gtsave(gt, "res/DescritoresSPP.png")
158
159 ###NORMALIDADE-----
160 library(moments)
161 Assimetria <- apply(m_trab,1,skewness)
162 Assimetria
163 Curtose <- apply(m_trab,1,kurtosis)
164 Curtose
165 Normalidade1 <- cbind(Assimetria, Curtose)
166 Normalidade1 <- as.data.frame(Normalidade1)
167 Normalidade1
168 SomaTotalN <- apply(Normalidade1,2,sum)
169 SomaTotalN
170 MediaTotalN <- apply(Normalidade1,2,mean)
171 MediaTotalN
172 DPTotalN <- apply(Normalidade1,2,sd)
173 DPTotalN
174 Normalidade2<-cbind(SomaTotalN, MediaTotalN, DPTotalN)
175 Normalidade2<-as.data.frame(Normalidade2)
176 Normalidade2 <- t(Normalidade2) # "t" transpoe a matriz
177 Normalidade2
178 NormalidadeFinal <- rbind(Normalidade1, Normalidade2)
179 NormalidadeFinal
180 NormalidadeFinal <- round(NormalidadeFinal, 2)
181 NormalidadeFinal
182

```

```

183 # Salvando NormalidadeFinal
184 write.table(data.frame("SPP"=rownames(NormalidadeFinal),
185                        NormalidadeFinal),
186            "res/NormalidadeSPP.txt",
187            row.names=FALSE,
188            sep="\t")
189
190 ###TABELA FINAL NORMALIDADE----
191 nf <- NormalidadeFinal
192 ncol(nf); nrow(nf) #no. de N colunas x M linhas
193 nf <- cbind(SPP = rownames(nf), nf)
194 gt <- gt(nf, rowname_col = "SPP", caption = "Descritores da normalidade por SPP (COLUNA)")
195 gt
196 gtsave(gt, "res/NormalidadeSPP.html")
197
198 #DIVERSIDADE EM DENSIDADE MATRIZ NÃO-TRANSPOSTA----
199
200 m_trab <- (m_dens) #HILL
201
202 # Números de Hill
203 Hill <- renyi(m_trab,
204             scales = c(0:5),
205             hill = TRUE)
206 Hill
207 library(ggplot2)
208 library(tidyr)
209 library(tidyverse)
210 grafico1 <- Hill %>%
211   rownames_to_column() %>%
212   pivot_longer(-rowname) %>%
213   mutate(name = factor(name, name[1:length(Hill)])) %>%
214   ggplot(aes(x = name, y = value, group = rowname,
215             col = rowname)) +
216   geom_point(size = 2) +
217   geom_line(size = 1) +
218   xlab("Parâmetro de ordem de diversidade (q)") +
219   ylab("Diversidade") +
220   labs(col = "Locais") +
221   theme_bw() +
222   theme(text = element_text(size = 10)) #ajustar a fonte caso nao caiba no output html
223 grafico1
224 ggsave(grafico1, dpi = 300, filename = "res/fig-hill.png")
225

```

```

226 # Gráfico de abundância
227 abund <- colSums(m_trab)
228 #abund <- m_trab[1, ] #escolhe a primeira linha para a distribuição de abundância
229 df <- data.frame(sp = colnames(m_trab),
230                 abun = abund)
231 grafico2 <- ggplot(df, aes(fct_reorder(sp, -abun),
232                                abun, group = 1)) +
233   geom_col() +
234   geom_line(col = "red", linetype = "dashed") +
235   geom_point(col = "red") +
236   xlab("Espécies") +
237   ylab("Abundância") +
238   theme_bw() +
239   theme(axis.text.x = element_text(
240     angle = 45,
241     hjust = 1,
242     face = "italic",
243     size = 10),
244     axis.text.y = element_text(size = 10),
245     axis.title.x = element_text(size = 12),
246     axis.title.y = element_text(size = 12))
247 grafico2
248 ggsave(grafico2, dpi = 300, filename = "res/fig-abun.png")
249
250 # Curvas de rarefação
251 m_trab_r <- mutate(as.data.frame(m_trab), across(everything(), ceiling)) #arredonda a matriz
252 colSums(t(m_trab_r))
253 colnames(t(m_trab_r))
254 #rarefa <- t(m_trab_r[c("S-R-CT2", "S-R-CP2", "S-A-TA2", "B-A-MU2"),]) #curva de rarefação para
255 class(m_trab)
256
257 # Curva de rarefação para todos os sítios
258 rarefa <- t(m_trab_r)
259 ## Curva de rarefação para as 8 UA's com maior soma
260 #m_trab_r <- as.data.frame(t(m_trab_r)) #transpõe a matriz
261 #col_sums <- colSums(m_trab_r)
262 #largest_columns <- names(sort(col_sums, decreasing = TRUE)[1:8])
263 #rarefa <- m_trab_r[largest_columns] #curva de rarefação para as 8 UA's com maior soma
264 library(iNEXT)
265 out <- iNEXT(rarefa, q = 0,
266             datatype = "abundance",
267             size = NULL,
268             endpoint = 2000, #define o comprimento de eixo x

```

```

269         knots = 40,
270         se = TRUE,
271         conf = 0.95,
272         nboot = 50)
273 grafico3 <- ggiNEXT(out, type = 1, facet.var="None") +
274   theme_bw() +
275   labs(fill = "Áreas") +
276   xlab("Número de indivíduos") +
277   ylab("Riqueza de espécies") +
278   theme(legend.title=element_blank()) #ver como fica com facet.var="Assemblage"
279 #grafico3
280 #ggsave(grafico3, dpi = 300, filename = "fig-rare1.png")
281
282 # Curvas de acumulação de espécies
283 acumula <- specaccum(m_trab,
284                     method = "random")
285 acumula
286 plot(acumula)
287 plot_data <- data.frame("UAs" = c(0, acumula$sites),
288                         "Riqueza" = c(0, acumula$richness),
289                         "lower" = c(0, acumula$richness - acumula$sd),
290                         "upper" = c(0, acumula$richness + acumula$sd))
291 gLocais <- ggplot(plot_data, aes(x = UAs, y = Riqueza)) +
292   geom_point(color = "blue", size = 4) +
293   geom_line(color = "blue", lwd = 2) +
294   geom_ribbon(aes(ymin = lower, ymax = upper),
295             linetype=2, alpha=0.3, fill = "yellow") +
296   ylab("Riqueza acumulada") +
297   theme_classic() +
298   theme(text = element_text(size = 16))
299 gLocais
300 ggsave(gLocais, dpi = 300, filename = "res/fig-acum.png")
301
302 plot_data <- data.frame("Individuos" = c(0, acumula$individuals),
303                         "Riqueza" = c(0, acumula$richness),
304                         "lower" = c(0, acumula$richness - acumula$sd),
305                         "upper" = c(0, acumula$richness + acumula$sd))
306 gInd <- ggplot(plot_data, aes(x = Individuos, y = Riqueza)) +
307   geom_point(color = "blue", size = 4) +
308   geom_line(color = "blue", lwd = 2) +
309   geom_ribbon(aes(ymin = lower, ymax = upper),
310             linetype=2, alpha=0.3, fill = "yellow") +
311   ylab("Riqueza acumulada") +

```

```

312   theme_classic() +
313   theme(text = element_text(size = 16))
314 gInd
315
316 #RIQUEZA RAREFACIONADA----
317
318 # Rarefação baseada na matriz de contagem
319 library(vegan)
320 data <- m_cont #raremax precisa de numeros inteiros
321 S <- specnumber(data) #observed number of species
322 S
323 #raremax <- min(rowSums(data))
324 raremax <- round(mean(rowSums(data)))
325 #raremax <- 1000
326 raremax
327 Srare <- rarefy(data, raremax)
328 Srare
329 plot(S, Srare, xlab = "Observed No. of Species", ylab = "Rarefied No. of Species")
330 abline(0, 1)
331
332 #png("res/fig-rarecurve.png")
333 rarecurve(data, step = 20,
334           sample = raremax, #linhas de rarefação
335           col = "blue", cex = 0.6,
336           xlab = "Sample size", ylab = "Taxa", xlim = c()) #c(0,5000)
337 #dev.off()
338
339 ##TABELA DA COMUNIDADE----
340
341 library(dplyr)
342 library(tidyr)
343 m_trab <- m_dens %>%
344   rename_with(~ gsub("_", ".", .))
345 m <- m_trab %>%
346   group_by(Rio = t_grps$Rio,
347            Periodo = t_grps$Periodo,
348            Ponto = t_grps$PontoN) %>%
349   summarise(across(where(is.numeric),
350                    list(mean = mean, sd = sd)),
351            .groups = 'drop') %>%
352   pivot_longer(
353     cols = -c(Rio, Periodo, Ponto),
354     names_to = c("Variable", ".value"),

```

```

355     names_sep = "_"
356   )
357 m
358 m <- as.data.frame(m)
359 m_wide <- m %>%
360   mutate(stat_string = ifelse(
361     Variable == "nome_da_variavel", #escolhe uma variável pra ter 3 casas decimais
362     paste0(round(mean, 3), " (", round(sd, 3), ")"),
363     paste0(round(mean, 1), " (", round(sd, 1), ")")
364   )) %>%
365   unite("Location", Rio, Período, Ponto, sep = "_") %>%
366   dplyr::select(Variable, Location, stat_string) %>% #dplyr dá conflito com MASS
367   pivot_wider(names_from = Location, values_from = stat_string)
368
369 m_wide
370 m_wide <- as.data.frame(m_wide)
371 m_wide
372
373 # Salvado m_wide
374 write.table(m_wide, "res/m_wide_spp.txt",
375             sep = ";", dec = ".", #"\\t",
376             row.names = TRUE,
377             quote = TRUE,
378             append = FALSE)
379 m_wide_spp <- read.table("res/m_wide_spp.txt",
380                          sep = ";", dec = ".",
381                          row.names = 1,
382                          header = TRUE,
383                          na.strings = NA)
384
385 # Exportando dados para Excel
386 library(openxlsx)
387 #write.xlsx(m_wide_spp, file = "res/tabela da comunidade.xlsx", rowNames = FALSE)
388 wb <- loadWorkbook("res/tabela da comunidade.xlsx")
389 writeData(wb, sheet = "Sheet 1", x = m_wide_spp)
390 saveWorkbook(wb, "res/tabela da comunidade.xlsx", overwrite = TRUE)
391
392 # Tabela final ajustada no Excel
393 library(readxl)
394 spp_gtable <- read_excel(
395   path = "res/tabela da comunidade.xlsx",
396   sheet = "tabela_final",
397   col_names = T,

```

```

398   range = "A1:AE61") #ajustar para o tamanho da tabela final
399 spp_gtttable
400 library(gt)
401 spp_gtttable <- gt(spp_gtttable)
402 spp_gtttable <- sub_missing(spp_gtttable,
403                             columns = everything(), missing_text = "")
404 #spp_gtttable <- fmt_number(spp_gtttable,
405 #                             columns = "F.O.(%)", decimals = 1)
406 spp_gtttable
407 gtsave(spp_gtttable, "res/gt-spp_gtttable_xlsx.html")
408 saveRDS(spp_gtttable, "res/gt-spp_gtttable_xlsx.rds")
409
410 #LISTA DE ESPÉCIES
411
412 # Exportando dados para Excel
413 library(openxlsx)
414 #write.xlsx(m_wide_spp, file = "res/tabela da comunidade.xlsx", rowNames = FALSE)
415 wb <- loadWorkbook("res/tabela da comunidade.xlsx")
416 writeData(wb, sheet = "Sheet 1", x = m_wide_spp)
417 saveWorkbook(wb, "res/tabela da comunidade.xlsx", overwrite = TRUE)
418
419 # Tabela final ajustada no Excel
420 library(readxl)
421 spp_gtttable <- read_excel(
422   path = "dados/rebio23-zoopl.xlsx",
423   sheet = "Espécies",
424   col_names = T,
425   range = "L1:S90") #ajustar para o tamanho da tabela final
426 spp_gtttable
427 library(gt)
428 spp_gtttable <- gt(spp_gtttable)
429 spp_gtttable <- sub_missing(spp_gtttable,
430                             columns = everything(), missing_text = "")
431 #spp_gtttable <- fmt_number(spp_gtttable,
432 #                             columns = "F.O.(%)", decimals = 1)
433 spp_gtttable
434 gtsave(spp_gtttable, "res/gt-checklist_gtttable_xlsx.html")
435 saveRDS(spp_gtttable, "res/gt-checklist_gtttable_xlsx.rds")
436
437 # Escolher sumário de uma variavel
438 m
439 var <- "h.roots"
440 m[m$Variable == var, "mean"] #cada valor de var

```

```

5  summary(m[m$Variable == var, "mean"]) #sumário dos valores de var
6
7  # Escolher sumário de um grupo de variáveis do df m
8  vars <- unique(grep("^h\\.\"", m$Variable, value = TRUE))
9  summaries <- list() #cria uma lista vazia para guardar os sumários
10 # Loop para cada variável do grupo e guarda em summaries
11 for (var in vars) {
12   summaries[[var]] <- summary(m[m$Variable == var, "mean"])
13 }
14 #var is a temporary variable used in the for loop to iterate through
15 #each variable name that starts with "h."
16
17 summaries
18 summary_table <- do.call(rbind, lapply(summaries, as.data.frame.list))
19 round(summary_table, 2)
20 #sink(file = "summary_h.txt", split = TRUE)
21 round(summary_table[order(summary_table$Mean, decreasing = FALSE), ], 2)
22 #sink()
23
24 # Tabela limpa
25 #summary_table <- cbind(Variable = rownames(summary_table), summary_table)
26 #rownames(summary_table) <- NULL
27 #colnames(summary_table) <- c("Variable", "Min", "Q1", "Median", "Mean", "Q3", "Max")
28 #summary_table

```

Rarefaction analysis showed that the Caiana stream had a taxa richness slightly higher than that of the Barro Branco stream (Figure 4). Nevertheless, given a similar sample size, the Barro Branco stream would have a greater diversity, as shown by the Chao index that estimated an extrapolated richness of 58 (± 8.9) species for the Barro Branco stream, compared with 54.2 (± 4.3) species for the Caiana stream.

```

1  #ZOOREBIO_part - CURVA DE ACUMULAÇÃO----
2  #####....-----
3
4  ##ORGANIZANDO DADOS----
5
6  dev.off() #apaga os graficos, se houver algum
7  rm(list=ls(all=TRUE)) #limpa a memória
8  cat("\014") #limpa o console
9
10 m_cont <- read.csv("res/m_cont.csv",
11                   sep = ";", dec = ".",
12                   row.names = 1,

```

```

13             header = TRUE,
14             na.strings = NA)
15 m_trab <- m_cont
16
17 library(tidyverse)
18
19 m_trab_r <- mutate(m_trab, across(everything(), ceiling)) #arredonda a matriz original para
20
21 colSums(t(m_trab_r))
22 colnames(t(m_trab_r))
23
24 ca_abund <- colSums(m_trab_r[grepl("-C-", rownames(m_trab_r)), ])
25 bb_abund <- colSums(m_trab_r[grepl("-B-", rownames(m_trab_r)), ])
26
27 m_rios <- rbind("Caiana" = ca_abund, "Barro Branco" = bb_abund)
28 m_rios
29
30 rarefa <- t(m_rios)
31
32 library(iNEXT)
33 out <- iNEXT(rarefa, q = 0,
34             datatype = "abundance",
35             size = NULL,
36             endpoint = 15000, #define o comprimento de eixo x
37             knots = 40,
38             se = TRUE,
39             conf = 0.95,
40             nboot = 50)
41
42 library(ggplot2)
43 grafico3 <- ggiNEXT(out, type = 1, facet.var="None") + #ver como fica com facet.var="Assembl
44   theme_bw() +
45   labs(fill = "Áreas") +
46   xlab("Number of individuals") +
47   ylab("Taxa richness") +
48   scale_linetype_discrete(
49     labels = c("Interpolation", "Extrapolation")) +
50   theme(legend.title = element_blank(),
51         legend.position = "bottom",
52         legend.text = element_text(size = 14),
53         axis.title = element_text(size = 14),
54         axis.text = element_text(size = 12)
55 )

```

```

56 grafico3
57 ggsave(grafico3, width = 8, height = 6, dpi = 300, filename = "res/fig-rare1.png")
58
59 #ESTIMADORES DA DIVERSIDADE
60
61 #https://rpubs.com/Juan-Mt/diversity-estimations
62 #https://analises-ecologicas.com/cap11
63
64 library(vegan)
65 m_cont <- as.matrix(m_cont)
66 storage.mode(m_cont) <- "numeric"
67
68 BB <- m_cont[grepl("^[SC]-B-", rownames(m_cont)),, drop = FALSE]
69 CA <- m_cont[grepl("^[SC]-C-", rownames(m_cont)),, drop = FALSE]
70
71 nrow(BB)
72 nrow(CA)
73
74 # Calculando a riqueza observada
75 BB_total <- colSums(BB)
76 CA_total <- colSums(CA)
77 S_BB <- sum(BB_total > 0)
78 S_CA <- sum(CA_total > 0)
79 S_BB
80 S_CA
81
82 # Chao1
83 chao_BB <- estimateR(BB_total)
84 chao_CA <- estimateR(CA_total)
85 chao_BB
86 chao_CA
87
88 # Separando Chao1
89 resultado_chao <- data.frame(
90   Stream = c("Barro Branco", "Caiana"),
91   Observed = c(
92     chao_BB["S.obs"],
93     chao_CA["S.obs"]
94   ),
95   Chao1 = c(
96     chao_BB["S.chao1"],
97     chao_CA["S.chao1"]
98   ),

```

```

99   SE = c(
100     chao_BB["se.chao1"],
101     chao_CA["se.chao1"]
102   )
103 )
104 resultado_chao
105
106 # 95% confidence intervals
107 resultado_chao$Lower95 <- resultado_chao$Chao1 -
108   1.96 * resultado_chao$SE
109 resultado_chao$Upper95 <- resultado_chao$Chao1 +
110   1.96 * resultado_chao$SE
111
112 # Do not allow the lower limit to be below observed richness
113 resultado_chao$Lower95 <- pmax(
114   resultado_chao$Lower95,
115   resultado_chao$Observed
116 )
117
118 sink(file = "res/chao.txt")
119 # Tabela pronta
120 resultado_chao
121 sink()

```

Regarding the community composition as shown by the UPGMA (Figure 5), sites did not aggregate to represent the different streams or their hierarchy across seasons. Even though the Cophenetic coefficients showed that the UPGMA dendrograms were a good representation of the original Bray-Curtis dissimilarities among sampling sites ($CC_{dry} = 0.84$ and $CC_{wet} = 0.87$), the UPGMA cluster analyses revealed low-to-moderate spatial clustering strength for the zooplankton communities across both periods, as indicated by the Agglomerative coefficients ($AC_{dry} = 0.34$ and $AC_{wet} = 0.39$). The PERMANOVA based on Bray-Curtis dissimilarities showed a significant effect of both hydrological period ($F_{1,86} = 5.3, R^2 = 0.056, p = 0.001$) and stream ($F_{1,86} = 1.8, R^2 = 0.019, p = 0.042$) on community composition. The full model including the Season $\times$ Stream interaction was also significant ($F_{1,86} = 2.1, R^2 = 0.022, p = 0.031$), explaining 9.73% of the total variation, with 90.27% of the variation unexplained by the factors included in the model.

```

1  #ZOOEBIO_part - UPGMA----
2  #####.....-----
3
4  ##ORGANIZANDO DADOS----
5
6  dev.off() #apaga os graficos, se houver algum

```

```

7  rm(list=ls(all=TRUE)) #limpa a memória
8  cat("\014") #limpa o console
9
10 t_grps <- read.csv("res/t_grps.csv",
11                    sep = ";", dec = ".",
12                    row.names = 1,
13                    header = TRUE,
14                    na.strings = NA)
15 m_dens <- read.csv("res/m_dens.csv",
16                   sep = ";", dec = ".",
17                   row.names = 1,
18                   header = TRUE,
19                   na.strings = NA)
20 m_hab_part <- read.csv("res/m_hab_part.csv",
21                       sep = ";", dec = ".",
22                       row.names = 1,
23                       header = TRUE,
24                       na.strings = NA)
25 m_cont <- read.csv("res/m_cont.csv",
26                   sep = ";", dec = ".",
27                   row.names = 1,
28                   header = TRUE,
29                   na.strings = NA)
30
31 #TRANSFORMANDO A MATRIZ DE DENSIDADE
32
33 library(vegan)
34 m_dens_trns <- asin(sqrt(decostand(m_dens,
35                                  method="total", MARGIN = 2)))
36
37 write.table(m_dens_trns, "res/m_dens_trns.csv",
38            sep = ";", dec = ".", #"\\t",
39            row.names = TRUE,
40            quote = TRUE,
41            append = FALSE)
42 m_dens_trns <- read.csv("res/m_dens_trns.csv",
43                       sep = ";", dec = ".",
44                       row.names = 1,
45                       header = TRUE,
46                       na.strings = NA)
47
48 vegdist <- vegdist(m_dens_trns, method = "bray",
49                   diag = TRUE,

```

```

50         upper = FALSE)
51 cluster <- hclust(vegdist, method = "average")
52 plot (cluster, main = "Cluster Dendrogram - Bray-Curtis")
53 rect.hclust(cluster, k = 3, h = NULL)
54 #h = 0.8 fornece os grupos formados na altura h
55 as.matrix(vegdist)[1:6, 1:6]
56
57 #SEPARANDO OS PERÍODOS E REFAZENDO A TRANSFORMAÇÃO
58
59 # Fazendo a média por ponto
60 ponto_id <- sub("-H[0-9]+$", "", rownames(m_dens))
61 m_dens_media <- aggregate(m_dens, by = list(Ponto = ponto_id), FUN = mean)
62 rownames(m_dens_media) <- m_dens_media$Ponto
63 m_dens_media$Ponto <- NULL
64
65 write.table(m_dens_media, "res/m_dens_media.csv",
66             sep = ";", dec = ".", #"\t",
67             row.names = TRUE,
68             quote = TRUE,
69             append = FALSE)
70 m_dens_media <- read.csv("res/m_dens_media.csv",
71                         sep = ";", dec = ".",
72                         row.names = 1,
73                         header = TRUE,
74                         na.strings = NA)
75 m_dens_media <- m_dens_trns
76 # Separar por período
77 seco_med <- m_dens_media[startsWith(rownames(m_dens_media), "S-"), ]
78 chuv_med <- m_dens_media[startsWith(rownames(m_dens_media), "C-"), ]
79 # Filtrar COLUNAS zeradas (espécies ausentes no período)
80 seco_med <- seco_med[, colSums(seco_med) > 0]
81 chuv_med <- chuv_med[, colSums(chuv_med) > 0]
82 # Filtrar LINHAS zeradas (pontos sem organismos)
83 seco_med <- seco_med[rowSums(seco_med) > 0, ]
84 chuv_med <- chuv_med[rowSums(chuv_med) > 0, ]
85 # Transformação
86 seco_trns <- asin(sqrt(decostand(seco_med,
87                                method="total", MARGIN = 2)))
88 chuv_trns <- asin(sqrt(decostand(chuv_med,
89                                method="total", MARGIN = 2)))
90 # Distâncias de Bray-Curtis e UPGMA
91 dist_seco <- vegdist(seco_trns, method = "bray")
92 dist_chuv <- vegdist(chuv_trns, method = "bray")

```

```

93 upgma_seco <- hclust(dist_seco, method = "average")
94 upgma_chuv <- hclust(dist_chuv, method = "average")
95
96 # Coeficientes de aglomeração
97 agg_coef_seco <- coef.hclust(upgma_seco)
98 agg_coef_chuv <- coef.hclust(upgma_chuv)
99 # Gerar a matriz de distância cofenética a partir da árvore UPGMA
100 cofe_seco <- cophenetic(upgma_seco)
101 cofe_chuv <- cophenetic(upgma_chuv)
102 # Calcular a correlação com a matriz de distância original (Bray-Curtis)
103 cor_seco <- cor(dist_seco, cofe_seco)
104 cor_chuv <- cor(dist_chuv, cofe_chuv)
105 # Visualizar os resultados
106 cat("Correlação Cofenética (Seco):", round(cor_seco, 3), "\n")
107 cat("Correlação Cofenética (Chuvoso):", round(cor_chuv, 3), "\n")
108
109 # Plot dos dendrogramas agrupados por ponto
110 upgma_seco$labels <- sub("[A-Z]-", "", upgma_seco$labels)
111 upgma_chuv$labels <- sub("[A-Z]-", "", upgma_chuv$labels)
112 dend_seco <- as.dendrogram(hclust(dist_seco, method = "average"))
113 dend_chuv <- as.dendrogram(hclust(dist_chuv, method = "average"))
114
115 #png("res/fig-upgma.png", width=11, height=9, units="in", res=400)
116 par(mfrow = c(1, 2), mar = c(5, 4, 2, 6)) #B, L, T, R
117 plot(as.dendrogram(upgma_seco),
118      horiz = TRUE,
119      xlab = "Bray-Curtis dissimilariry",
120      xlim = c(max(attr(dend_seco, "height")), 0),
121      cex.lab = 1)
122 title(main = "Dry", line = 0, cex.main = 1.2)
123 mtext(paste0("Agglomerative Coefficient = ", round(agg_coef_seco, 2)),
124      side = 2,           # Lado esquerdo (eixo Y)
125      line = 1,           # Distância da margem
126      at = 7,             # Posição na base do eixo
127      las = 0,            # Orientação do texto
128      cex = 0.8)          # Tamanho da fonte
129 mtext(paste0("Cophenetic Coefficient = ", round(cor_seco, 2)),
130      side = 2,           # Lado esquerdo (eixo Y)
131      line = 2,           # Distância da margem
132      at = 7,             # Posição na base do eixo
133      las = 0,            # Orientação do texto
134      cex = 0.8)          # Tamanho da fonte
135

```

```

136 plot(as.dendrogram(upgma_chuv),
137       horiz = TRUE,
138       xlab = "Bray-Curtis dissimilariry",
139       xlim = c(max(attr(dend_chuv, "height")), 0),
140       cex.lab = 1)
141 title(main = "Wet", line = 0, cex.main = 1.2)
142 mtext(paste0("Agglomerative Coefficient = ", round(agg_coef_chuv, 2)),
143       side = 2,           # Lado esquerdo (eixo Y)
144       line = 1,           # Distância da margem
145       at = 7,             # Posição na base do eixo
146       las = 0,            # Orientação do texto
147       cex = 0.8)          # Tamanho da fonte
148 mtext(paste0("Cophenetic Coefficient = ", round(cor_chuv, 2)),
149       side = 2,           # Lado esquerdo (eixo Y)
150       line = 2,           # Distância da margem
151       at = 7,             # Posição na base do eixo
152       las = 0,            # Orientação do texto
153       cex = 0.8)          # Tamanho da fonte
154 par(mfrow = c(1, 1))
155 #dev.off()
156
157 #OUTRA FORMA
158 #plot(as.dendrogram(upgma_chuv),
159 #      # horiz = TRUE,
160 #      # main = "Wet",
161 #      # xlab = "Bray-Curtis dissimilarity",
162 #      # xlim = c(max(attr(dend_chuv, "height")), 0))
163 # Adiciona o texto no canto inferior esquerdo
164 #text(x = max(attr(dend_chuv, "height")) * 0.9,
165 #      # y = 3,
166 #      # labels = paste0("Agglomerative Coefficient = ", round(agg_coef_chuv, 2)),
167 #      # adj = c(0, 0),
168 #      # cex = 0.85,
169 #      # srt = 90) # srt = 90 gira o texto para ficar vertical
170 #OU
171 #mtext(paste0("Cophenetic Coefficient = ", round(cor_seco, 2)),
172 #      # side = 2,      # 1 = bottom margin
173 #      # line = 2.5,    # Line distance below axis (increase if it overlaps numbers)
174 #      # adj = 0.36,    # Posição horizontal
175 #      # cex = 0.85)    # Font size

1 #ORGANIZANDO DADOS----
2

```

```

3 dev.off()
4 rm(list=ls(all=TRUE))
5 cat("\014")
6
7 t_grps <- read.csv("res/t_grps.csv",
8                   sep = ";", dec = ".",
9                   row.names = 1,
10                  header = TRUE,
11                  na.strings = NA)
12 m_dens_trns <- read.csv("res/m_dens_trns.csv",
13                        sep = ";", dec = ".",
14                        row.names = 1,
15                        header = TRUE,
16                        na.strings = NA)
17
18 colnames(t_grps)
19 m_trab <- m_dens_trns
20 library(vegan)
21
22 #MRPP----
23
24 #TRABALHANDO COM MULTIPLAS VARIAVEIS----
25
26 # Run MRPP for multiple variables
27 fatores <- c("Periodo", "Rio", "Ref_site")
28 set.seed(123)
29 mrpp_results <- lapply(fatores, function(v) {
30   grp <- as.factor(t_grps[[v]])
31   mrpp(
32     dat = m_trab,
33     grouping = grp,
34     distance = "bray"
35   )
36 })
37 names(mrpp_results) <- fatores
38 mrpp_results
39 # Print one result
40 mrpp_results$Ref_site
41 # Loop through summaries
42 sink(file = "res/mrpp-dens.txt", split = TRUE)#, append = TRUE)
43 for (v in names(mrpp_results)) {
44   cat("\n=====\\n")
45   cat("MRPP for:", v, "\\n")

```

```

46   print(mrpp_results[[v]])
47 }
48 sink()
49 # Extract key statistics into a table
50 mrpp_summary <- data.frame(
51   Variable = fatores,
52   Delta = sapply(mrpp_results, function(x) x$delta),
53   Expected_Delta = sapply(mrpp_results, function(x) x$E.delta),
54   A = sapply(mrpp_results, function(x) x$A),
55   P_value = sapply(mrpp_results, function(x) x$Pvalue)
56 )
57 sink("res/mrpp-dens.txt", append = TRUE)
58 mrpp_summary
59 sink()
60
61 ##TRABALHANDO COM UMA UNICA VARIABEL----
62
63 # Run MRPP for a single variable
64 grp <- as.factor(t_grps$Ref_site) #UA, mes, estacao, area, ambiente
65 set.seed(123)
66 mrpp <- mrpp(dat = m_trab, grouping = grp, distance = "bray")
67 mrpp
68 summary(mrpp)
69
70 # Teste de hipótese para uma unica variavel
71 str(mrpp)
72 p_value <- mrpp$Pvalue
73 print(sprintf("p-value: %.10f", p_value))
74 # Conditional statement to check the p-value and print the appropriate message
75 if (p_value < 0.05) {
76   print(sprintf("Existe diferença significativa porque o valor de p é %.10f, sendo menor que
77 } else {
78   print(sprintf("Não existe diferença significativa porque o valor de p é %.10f, sendo maior
79 }
80
81 ##TESTES PAREADOS----
82
83 # Todos os pares de níveis do fator
84 pairs <- combn(levels(as.factor(grp)), 2, simplify = FALSE)
85 pairs
86 # Aplicar mrpp para cada par
87 res_list <- lapply(pairs, function(p) {
88   sel <- grp %in% p

```

```

89   mrpp(dat = m_trab[sel, ], grouping = droplevels(grp[sel]), distance = "bray")
90 })
91 # Dar nomes aos resultados
92 names(res_list) <- sapply(pairs, function(p) paste(p, collapse = "_vs_"))
93 # visualizar
94 res_list
95 # Resumo
96 summary_tab <- data.frame(
97   Comparison = names(res_list),
98   A = sapply(res_list, function(x) x$A),
99   Delta = sapply(res_list, function(x) x$delta),
100   Pvalue = sapply(res_list, function(x) x$Pvalue)
101 )
102 sink("res/mrpp-dens.txt", append = TRUE)
103 summary_tab
104 sink()
105
106 ###MATRIX-STYLE TABLE----
107 library(dplyr)
108 library(tidyr)
109 library(knitr)
110 library(kableExtra)
111 # Extract unique group names from the comparisons
112 groups1 <- sort(unique(unlist(strsplit(summary_tab$Comparison, "_vs_"))))
113 # Initialize empty matrix
114 pval_matrix <- matrix(NA, nrow = length(groups1), ncol = length(groups1),
115                       dimnames = list(groups1, groups1))
116 # Fill matrix with Pvalues
117 for (i in 1:nrow(summary_tab)) {
118   comp <- unlist(strsplit(summary_tab$Comparison[i], "_vs_"))
119   g1 <- comp[1]
120   g2 <- comp[2]
121   p <- summary_tab$Pvalue[i]
122
123   pval_matrix[g1, g2] <- p
124   pval_matrix[g2, g1] <- p #make symmetric
125 }
126
127 # Format p-values: red if <0.05
128 pval_colored <- ifelse(is.na(pval_matrix), "",
129                        ifelse(pval_matrix < 0.05,
130                               paste0("<span style='color:red;'>", sprintf("%.3f", pval_matrix[i, j])),
131                               sprintf("%.3f", pval_matrix[i, j]))

```

```

132
133 # Convert to data.frame for kable
134 pval_df <- as.data.frame(pval_colored)
135 pval_df <- tibble::rownames_to_column(pval_df, "Group")
136 # Render HTML table
137 matrixstyle <- pval_df %>%
138   kable("html", escape = FALSE) %>%
139   kable_styling(full_width = FALSE)
140 matrixstyle
141 save_kable(matrixstyle, "res/kb-mrpp-dens_matrixstyle.html")
142
143 ##GRÁFICO DA MRPP----
144
145 png("res/fig-mrpp-dens.png", width = 15, height = 10, units = "in", res = 300)
146 def.par <- par(no.readonly = TRUE)
147 layout(matrix(1:2,nr=1))
148 plot(ord <- metaMDS(m_trab, distance="bray"), type="text", display="sites")
149 with(t_grps, ordihull(ord, grp))
150 with(mrpp, {
151   fig.dist <- hist(boot.deltas, xlim=range(c(delta,boot.deltas)),
152                   main="Test of Differences Among Groups")
153   abline(v=delta);
154   text(delta, 2*mean(fig.dist$counts), adj = -0.5,
155         expression(bold(delta)), cex=1.5 ) }
156 )
157 par(def.par)
158 dev.off()
159
160 ###AVALIANDO AS DISTÂNCIAS MÉDIAS-----
161 md <- with(t_grps, meandist(vegdist(m_trab), grp))
162 md
163 summary(md)
164 par(mfrow=c(1,2))
165 plot(md)
166 plot(md, kind="histogram")
167 par(mfrow=c(1,1))
168
169 #PerMANOVA SIMPLIFICADA----
170
171 #browseURL("https://riffomonas.org/code_club/2021-03-18-vegan")
172 #browseURL("https://riffomonas.org/code_club/2021-03-22-reproducibility")
173
174 library(vegan)

```

```

175
176 ## TESTES PARA VARIÁVEIS COM MAIS DE DOIS FATORES
177
178 adonis2(m_trab ~ Ref_site, data = t_grps, method = "bray")
179 set.seed(123)
180 perm <- 999
181 all_test <- adonis2(m_trab ~ Ref_site, data = t_grps, method = "bray", permutations = perm)
182 all_test
183
184 ## TESTES PARA VARIÁVEIS COM DOIS FATORES
185
186 sink(file = "res/perma-dens.txt", split = TRUE) #, append = TRUE)
187 set.seed(123)
188 perm <- 999
189 adonis2(m_trab ~ Período, data = t_grps, method = "bray", permutations = perm)
190 adonis2(m_trab ~ Rio, data = t_grps, method = "bray", permutations = perm)
191 adonis2(m_trab ~ Período + Rio, data = t_grps, #+ no lugar de * tira os resíduos
192         method = "bray", permutations = perm)
193
194 ## AVALIANDO INTERAÇÕES-----
195
196 adonis2(m_trab ~ Período*Rio, data = t_grps, method = "bray", permutations = perm,
197        by = "terms") #ou by=NULL remove as interações
198 adonis2(m_trab ~ Período*Rio, data = t_grps, method = "bray", permutations = perm,
199        by = "margin") #ou by=NULL remove as interações
200 adonis2(m_trab ~ Período*Rio, data = t_grps, method = "bray", permutations = perm,
201        by = NULL) #ou by=NULL remove as interações
202 sink()
203
204 #BETADISPER
205
206 dist_bray <- vegdist(m_trab, method = "bray")
207 t_grps$Período_Rio <- interaction(
208     t_grps$Período,
209     t_grps$Rio,
210     sep = "_")
211 bd <- betadisper(dist_bray, t_grps$Período_Rio)
212 permutest(bd, permutations = 999)
213
214 # PARA CADA FATOR
215
216 bd_período <- betadisper(dist_bray, t_grps$Período)
217 bd_rio <- betadisper(dist_bray, t_grps$Rio)

```

```

218 permutest(bd_periodo, permutations = 999)
219 permutest(bd_rio, permutations = 999)

```

Hierarchical additive partitioning was firstly performed separate for each season with the gamma diversity including both study stream. The comparison between dry and wet seasons revealed that zooplankton diversity was structured non-randomly across spatial scales ($p < 0.001$ for all components except β_2 ; Table 4). For the dry season, the alpha diversity ($\alpha_1 = 12.5\%$) was significantly lower than expected by chance ($SES = -22.1, p = 0.001$), indicating spatial aggregation or environmental filtering at finer scales. Habitat-scale beta diversity ($\beta_1 = 9.7\%$) was significantly lower than expected under the null model ($SES = -8.9, p = 0.001$), reflecting high compositional similarity among local sampling units. On the other hand, beta diversity, among streams ($\beta_3 = 26.7\%$) was significantly higher than that predicted by chance ($SES = 5.6, p = 0.001$), demonstrating substantial species turnover at the highest hierarchical level. The intermediate scale (among sites) contributed the most for the observed regional diversity ($\beta_2 = 51.1\%$) and it did not deviate significantly from null expectations ($SES = 0.6, p = 0.519$). Contrasting results were obtained for the wet season, when the observed alpha diversity ($\alpha_1 = 23.7\%$) was much higher than the dry season, yet significantly lower than that expected by chance ($SES = -45.4, p = 0.001$). Similar to results from the dry season, the habitat-scale beta diversity ($\beta_1 = 11.8\%$) was significantly lower than expected under the null model ($SES = -7.0, p = 0.001$), and the the beta diversity among streams ($\beta_3 = 16.3\%$) was significantly higher than that predicted by chance ($SES = 9.5, p = 0.001$). The intermediate scale (among sites) gave a lesser contribution to the total observed diversity ($\beta_2 = 48.3\%$), when compared to the dry season and, differently from the dry season, it deviated significantly from the null expectation ($SES = 13.4, p = 0.001$).

Hierarchical additive partitioning was performed separately for each season within the Barro Branco and Caiana streams (Figure 7). Across all four combinations, local alpha diversity ($\alpha_1 = 14.7\% - 30.8\%$) and fine-scale habitat beta diversity ($\beta_1 = 12.1\% - 14.8\%$) were consistently lower than expected under null model predictions (see `?@tbl-part_gtttables` for SES and P -values). The alpha diversity was higher during the wet season when compared with the dry season across both streams. This recurring negative deviation highlights strong local spatial aggregation and compositional homogenization among sampling points within sites. Beta diversity among sites ($\beta_2 = 54.6\% - 73.2\%$) consistently contributed the largest share of total regional richness and was significantly higher than predicted by chance (see `?@tbl-part_gtttables` for SES and P -values). These positive effect sizes confirm that spatial species turnover among sites is the primary driver structuring zooplankton assemblages within both streams regardless of seasonality.

```

1 #ZOOEBIO_part - ADDPART----
2 #####....----
3
4 ##ORGANIZANDO DADOS----
5
6 dev.off() #apaga os graficos, se houver algum

```

```

7  rm(list=ls(all=TRUE)) #limpa a memória
8  cat("\014") #limpa o console
9
10 m_esc <- read.csv("res/m_esc.csv",
11                  sep = ";", dec = ".",
12                  row.names = 1,
13                  header = TRUE,
14                  na.strings = NA)
15 m_dens <- read.csv("res/m_dens.csv",
16                  sep = ";", dec = ".",
17                  row.names = 1,
18                  header = TRUE,
19                  na.strings = NA)
20 m_cont <- read.csv("res/m_cont.csv",
21                  sep = ";", dec = ".",
22                  row.names = 1,
23                  header = TRUE,
24                  na.strings = NA)
25
26 #BUSCANDO AS ESCALAS
27
28 # Partição total
29 colnames(m_esc)
30 pt.total <- m_esc[, c("pt.rio", "pt.periodo", "pt.rebio")]
31 head(pt.total)
32 dim(pt.total)
33
34 # Partição seco
35 pt.seco <- m_esc[, c("s.habitat", "s.ponto", "s.rio", "s.rebio")]
36 pt.seco <- pt.seco[grepl("^S-", rownames(pt.seco)), ]
37 head(pt.seco)
38 dim(pt.seco)
39 m_cont_s <- m_cont[grepl("^S-", rownames(m_cont)), ]
40
41 # Partição chuvoso
42 pt.chuva <- m_esc[, c("c.habitat", "c.ponto", "c.rio", "c.rebio")]
43 pt.chuva <- pt.chuva[grepl("^C-", rownames(pt.chuva)), ]
44 head(pt.chuva)
45 dim(pt.chuva)
46 m_cont_c <- m_cont[grepl("^C-", rownames(m_cont)), ]
47
48 # Partição bb.seco (A)
49 bb.seco <- m_esc[, c("bb.habitat", "bb.ponto", "bb.rio")]

```

```

50 bb.seco <- bb.seco[grepl("S-B-", rownames(bb.seco)), ]
51 head(bb.seco)
52 dim(bb.seco)
53 m_cont_bb.seco <- m_cont[grepl("S-B-", rownames(m_cont)), ]
54
55 # Partição bb.chuva (B)
56 bb.chuva <- m_esc[, c("bb.habitat", "bb.ponto", "bb.rio")]
57 bb.chuva <- bb.chuva[grepl("C-B-", rownames(bb.chuva)), ]
58 head(bb.chuva)
59 dim(bb.chuva)
60 m_cont_bb.chuva <- m_cont[grepl("C-B-", rownames(m_cont)), ]
61
62 # Partição ca.seco (C)
63 ca.seco <- m_esc[, c("ca.habitat", "ca.ponto", "ca.rio")]
64 ca.seco <- ca.seco[grepl("S-C-", rownames(ca.seco)), ]
65 head(ca.seco)
66 dim(ca.seco)
67 m_cont_ca.seco <- m_cont[grepl("S-C-", rownames(m_cont)), ]
68
69 # Partição ca.chuva (D)
70 ca.chuva <- m_esc[, c("ca.habitat", "ca.ponto", "ca.rio")]
71 ca.chuva <- ca.chuva[grepl("C-C-", rownames(ca.chuva)), ]
72 head(ca.chuva)
73 dim(ca.chuva)
74 m_cont_ca.chuva <- m_cont[grepl("C-C-", rownames(m_cont)), ]
75
76 #PARTIÇÃO
77
78 #pt.total (m_cont)
79 #pt.seco (m_cont_s), pt.chuva (m_cont_c)
80 #bb.seco(A) (m_cont_bb.seco), bb.chuva(B) (m_cont_bb.chuva)
81 #ca.seco(C) (m_cont_ca.seco), ca.chuva(D) (m_cont_ca.chuva)
82
83 library(vegan)
84 part1 <- ca.chuva
85 cont <- m_cont_ca.chuva
86 sink(file = "res/part-pt.ca.chuva.txt")
87 #VALORES ABSOLUTOS
88 set.seed(666)
89 part.a <- adpart(y = cont, x = part1, index = "richness", relative = FALSE,
90                 weights = "unif", nsimul = 999)
91 part.a
92

```

```

93  #EM PORCENTAGEM
94  set.seed(666)
95  part.p <- adipart(y = cont, x = part1, index = "richness", relative = TRUE,
96                  weights = "unif", nsimul = 999)
97  part.p
98
99  #saveRDS(part.p, "res/part.p-pt.total.rds")
100 #saveRDS(part.p, "res/part.p-pt.seco.rds")
101 #saveRDS(part.p, "res/part.p-pt.chuva.rds")
102 #saveRDS(part.p, "res/part.p-bb.seco.rds")
103 #saveRDS(part.p, "res/part.p-bb.chuva.rds")
104 #saveRDS(part.p, "res/part.p-ca.seco.rds")
105 #saveRDS(part.p, "res/part.p-ca.chuva.rds")
106
107 #part.p <- readRDS("res/part.p-pt.total.rds")
108 #part.p <- readRDS("res/part.p-pt.seco.rds")
109 #part.p <- readRDS("res/part.p-pt.chuva.rds")
110 #part.p <- readRDS("res/part.p-bb.seco.rds")
111 #part.p <- readRDS("res/part.p-bb.chuva.rds")
112 #part.p <- readRDS("res/part.p-ca.seco.rds")
113 #part.p <- readRDS("res/part.p-ca.chuva.rds")
114
115 # Extrair valores do objeto part.p
116 part2 <- part.p
117
118 statistic <- part2$statistic
119 SES <- part2$oecosimu$z
120 mean <- part2$oecosimu$means
121 lower_2.5 <- apply(part2$oecosimu$simulated, 1, quantile, probs = 0.025)
122 median_50 <- apply(part2$oecosimu$simulated, 1, quantile, probs = 0.50)
123 upper_97.5 <- apply(part2$oecosimu$simulated, 1, quantile, probs = 0.975)
124 Pr_sim <- part2$oecosimu$pval
125
126 # Definir função para adicionar significância
127 get_significance <- function(p) {
128   if (p <= 0.001) {
129     return("****")
130   } else if (p <= 0.01) {
131     return("**")
132   } else if (p <= 0.05) {
133     return("*")
134   } else if (p <= 0.1) {
135     return(".")
136   }
137 }

```

```

136   } else {
137     return(" ")
138   }
139 }
140 significance <- sapply(Pr_sim, get_significance)
141
142 # Criar dataframe com todas as informações
143 diversity_data <- data.frame(
144   Component = names(statistic),
145   Observed = round(statistic, 2),
146   SES = round(SES, 2),
147   Mean = round(mean, 2),
148   Lower_2.5 = round(lower_2.5, 2),
149   Median_50 = round(median_50, 2),
150   Upper_97.5 = round(upper_97.5, 2),
151   Pr_sim = Pr_sim,
152   Sign = significance
153 )
154
155 # Exibir a tabela
156 library(gt)
157 print(diversity_data)
158 #str(part2)
159 gt(diversity_data)
160
161 # Criar dataframe com valores observados e esperados
162 observed_values <- part2$statistic
163 observed_values
164 expected_values <- part2$oecosimu$means
165 expected_values
166
167 # Combine observed and expected values into a dataframe
168 df <- data.frame(
169   Component = rep(names(observed_values), 2), # Use names from observed_values
170   Type = rep(c("Observed", "Expected"), each = length(observed_values)),
171   Value = c(observed_values, expected_values) # Observado primeiro, Esperado depois
172 )
173
174 df$Type <- factor(df$Type, levels = c("Observed", "Expected"))
175 df
176 df_filtered <- df[!df$Component %in% c("alpha.2", "alpha.3", "alpha.4", "gamma"), ]
177 df_filtered$Component <- factor(df_filtered$Component, levels = c("beta.4", "beta.3", "beta.2", "beta.1"))
178 df_filtered

```

```

179
180 diversity_data <- df_filtered
181 diversity_data
182
183 # Calcular porcentagens para cada componente
184 diversity_data <- diversity_data %>%
185   group_by(Type) %>%
186   mutate(Percentage = Value / sum(Value) * 100)
187
188 hier <- hiersimu(y = cont, x = part1, FUN = diversity)
189 hier
190 sink()
191
192 #saveRDS(diversity_data, "res/part-pt.gtotal.rds")
193 #saveRDS(diversity_data, "res/part-pt.gseco.rds")
194 #saveRDS(diversity_data, "res/part-pt.gchuva.rds")
195 #saveRDS(diversity_data, "res/part-bb.gseco.rds")
196 #saveRDS(diversity_data, "res/part-bb.gchuva.rds")
197 #saveRDS(diversity_data, "res/part-ca.gseco.rds")
198 #saveRDS(diversity_data, "res/part-ca.gchuva.rds")
199
200 #GRÁFICOS
201
202 library(ggplot2)
203 library(dplyr)
204
205 # PARTIÇÃO TOTAL
206
207 pt.gtotal <- readRDS("res/part-pt.gtotal.rds")
208 ggplot_pt.total <- ggplot(pt.gtotal, aes(fill = Component, y = Percentage, x = Type)) +
209   geom_bar(position = "stack", stat = "identity") +
210   labs(title = "Total",
211        y = "% of the total diversity",
212        x = "") +
213   scale_fill_manual(values = c("alpha.1" = "black",
214                                "beta.1" = "darkgray",
215                                "beta.2" = "gray",
216                                "beta.3" = "lightgray",
217                                "beta.4" = gray(0.9)),
218                     labels = c("alpha.1" = "Habitat ( 1)",
219                                "beta.1" = "Habitat ( 1)",
220                                "beta.2" = "Site ( 2)",
221                                "beta.3" = "Stream ( 3)",

```

```

222         "beta.4" = " 4")) +
223   theme_minimal() +
224   theme(
225     axis.text.x = element_text(angle = 0, hjust = 0.5),
226     panel.grid = element_blank(),
227     legend.position = "right"
228   )
229 ggplot_pt.total
230 ggsave(ggplot_pt.total, dpi = 300, filename = "res/fig-part_total.png")
231
232 # PARTIÇÃO POR PERÍODO
233
234 # Partição seco #pt.seco (m_cont_s)
235 pt.gseco <- readRDS("res/part-pt.gseco.rds")
236 ggplot_pt.seco <- ggplot(pt.gseco, aes(fill = Component, y = Percentage, x = Type)) +
237   geom_bar(position = "stack", stat = "identity") +
238   labs(title = "Dry",
239        y = "% of the total diversity",
240        x = "") +
241   scale_fill_manual(values = c("alpha.1" = "black",
242                                "beta.1" = "darkgray",
243                                "beta.2" = "gray",
244                                "beta.3" = "lightgray",
245                                "beta.4" = gray(0.9)),
246                     labels = c("alpha.1" = "Habitat ( 1)",
247                                "beta.1" = "Habitat ( 1)",
248                                "beta.2" = "Site ( 2)",
249                                "beta.3" = "Stream ( 3)",
250                                "beta.4" = " 4")) +
251   theme_minimal() +
252   theme(
253     plot.title = element_text(hjust = 0.5), #hjust 0=left, 0.5=center, 1=right
254     axis.text.x = element_text(angle = 0, hjust = 0.5),
255     panel.grid = element_blank(),
256   )
257 ggplot_pt.seco
258 ggsave(ggplot_pt.total, dpi = 300, filename = "res/fig-part_seco.png")
259
260 # Partição chuvoso #pt.chuva (m_cont_c)
261 pt.gchuva <- readRDS("res/part-pt.gchuva.rds")
262 ggplot_pt.chuva <- ggplot(pt.gchuva, aes(fill = Component, y = Percentage, x = Type)) +
263   geom_bar(position = "stack", stat = "identity") +
264   labs(title = "Wet",

```

```

265     y = "% of the total diversity",
266     x = "") +
267     scale_fill_manual(values = c("alpha.1" = "black",
268                                "beta.1" = "darkgray",
269                                "beta.2" = "gray",
270                                "beta.3" = "lightgray",
271                                "beta.4" = gray(0.9)),
272                        labels = c("alpha.1" = "Habitat ( 1)",
273                                "beta.1" = "Habitat ( 1)",
274                                "beta.2" = "Site ( 2)",
275                                "beta.3" = "Stream ( 3)",
276                                "beta.4" = " 4")) +
277     theme_minimal() +
278     theme(
279       plot.title = element_text(hjust = 0.5), #hjust 0=left, 0.5=center, 1=right
280       axis.text.x = element_text(angle = 0, hjust = 0.5),
281       panel.grid = element_blank(),
282     )
283 ggplot_pt.chuva
284 ggsave(ggplot_pt.total, dpi = 300, filename = "res/fig-part_chuva.png")
285
286 # PARTIÇÃO POR RIO
287
288 # bb.seco (A) #bb.seco(A) (m_cont_bb.seco)
289 bb.gseco <- readRDS("res/part-bb.gseco.rds")
290 ggplot_bb.seco <- ggplot(bb.gseco, aes(fill = Component, y = Percentage, x = Type)) +
291   geom_bar(position = "stack", stat = "identity") +
292   labs(title = "Dry",
293        y = "% of the total diversity",
294        x = "") +
295   scale_fill_manual(values = c("alpha.1" = "black",
296                                "beta.1" = "darkgray",
297                                "beta.2" = "gray",
298                                "beta.3" = "lightgray",
299                                "beta.4" = gray(0.9)),
300                       labels = c("alpha.1" = "Habitat ( 1)",
301                                "beta.1" = "Habitat ( 1)",
302                                "beta.2" = "Site ( 2)",
303                                "beta.3" = "Stream ( 3)",
304                                "beta.4" = " 4")) +
305   theme_minimal() +
306   theme(
307     plot.title = element_text(hjust = 0.5), #hjust 0=left, 0.5=center, 1=right

```

```

308     axis.text.x = element_text(angle = 0, hjust = 0.5),
309     panel.grid = element_blank(),
310 )
311 ggplot_bb.seco
312 ggsave(ggplot_bb.seco, dpi = 300, filename = "res/fig-bb.seco.png")
313
314 # bb.chuva (B) #bb.chuva(B) (m_cont_bb.chuva)
315 bb.gchuva <- readRDS("res/part-bb.gchuva.rds")
316 ggplot_bb.chuva <- ggplot(bb.gchuva, aes(fill = Component, y = Percentage, x = Type)) +
317   geom_bar(position = "stack", stat = "identity") +
318   labs(title = "Wet",
319        y = "Barro Branco stream",
320        x = "") +
321   scale_y_continuous(position = "right") +
322   scale_fill_manual(values = c("alpha.1" = "black",
323                                "beta.1" = "darkgray",
324                                "beta.2" = "gray",
325                                "beta.3" = "lightgray",
326                                "beta.4" = gray(0.9)),
327                    labels = c("alpha.1" = "Habitat (1)",
328                               "beta.1" = "Habitat (1)",
329                               "beta.2" = "Site (2)",
330                               "beta.3" = "Stream (3)",
331                               "beta.4" = "4")) +
332   theme_minimal() +
333   theme(
334     plot.title = element_text(hjust = 0.5), #hjust 0=left, 0.5=center, 1=right
335     axis.text.x = element_text(angle = 0, hjust = 0.5),
336     axis.text.y = element_blank(),
337     panel.grid = element_blank(),
338   )
339 ggplot_bb.chuva
340 ggsave(ggplot_bb.chuva, dpi = 300, filename = "res/fig-bb.chuva.png")
341
342 # ca.seco (C) #ca.seco(C) (m_cont_ca.seco)
343 ca.gseco <- readRDS("res/part-ca.gseco.rds")
344 ggplot_ca.seco <- ggplot(ca.gseco, aes(fill = Component, y = Percentage, x = Type)) +
345   geom_bar(position = "stack", stat = "identity") +
346   labs(title = "",
347        y = "% of the total diversity", #\n quebra de linha
348        x = "") +
349   scale_fill_manual(values = c("alpha.1" = "black",
350                                "beta.1" = "darkgray",

```

```

351         "beta.2" = "gray",
352         "beta.3" = "lightgray",
353         "beta.4" = gray(0.9)),
354     labels = c("alpha.1" = "Habitat (1)",
355               "beta.1" = "Habitat (1)",
356               "beta.2" = "Site (2)",
357               "beta.3" = "Stream (3)",
358               "beta.4" = "4")) +
359   theme_minimal() +
360   theme(
361     axis.text.x = element_text(angle = 0, hjust = 0.5),
362     panel.grid = element_blank(),
363   )
364 ggplot_ca.seco
365 ggsave(ggplot_ca.seco, dpi = 300, filename = "res/fig-ca.seco.png")
366
367 # ca.chuva (D) #ca.chuva(D) (m_cont_ca.chuva)
368 ca.gchuva <- readRDS("res/part-ca.gchuva.rds")
369 ggplot_ca.chuva <- ggplot(ca.gchuva, aes(fill = Component, y = Percentage, x = Type)) +
370   geom_bar(position = "stack", stat = "identity") +
371   labs(title = "",
372        y = "Caiana stream",
373        x = "") +
374   scale_y_continuous(position = "right") +
375   scale_fill_manual(values = c("alpha.1" = "black",
376                                "beta.1" = "darkgray",
377                                "beta.2" = "gray",
378                                "beta.3" = "lightgray",
379                                "beta.4" = gray(0.9)),
380                     labels = c("alpha.1" = "Habitat (1)",
381                               "beta.1" = "Habitat (1)",
382                               "beta.2" = "Site (2)",
383                               "beta.3" = "Stream (3)",
384                               "beta.4" = "4")) +
385   theme_minimal() +
386   theme(
387     axis.text.x = element_text(angle = 0, hjust = 0.5),
388     axis.text.y = element_blank(),
389     panel.grid = element_blank(),
390   )
391 ggplot_ca.chuva
392 ggsave(ggplot_ca.chuva, dpi = 300, filename = "res/fig-ca.chuva.png")
393

```

```

394 # PAINEL DE GRÁFICOS CHUVOSO E SECO
395
396 library(gridExtra)
397
398 # Organizar os gráficos em 2 linhas e 2 colunas
399 painel_c_s <- grid.arrange(
400   ggplot_pt.seco, ggplot_pt.chuva,
401   ncol = 1, nrow = 2
402 )
403
404 # Salvar o painel em alta resolução (300 DPI)
405 ggsave(
406   filename = "res/fig-painel_pt.cs.png",
407   plot = painel_c_s,
408   width = 4,
409   height = 8,
410   dpi = 300
411 )
412
413 # PAINEL DE GRÁFICOS POR RIO
414
415 library(patchwork)
416
417 shared_scale <- scale_fill_manual(
418   values = c("alpha.1" = "black",
419             "beta.1" = "darkgray",
420             "beta.2" = "gray",
421             "beta.3" = "lightgray",
422             "beta.4" = gray(0.9)),
423   labels = c("alpha.1" = "Habitat (1)",
424             "beta.1" = "Habitat (1)",
425             "beta.2" = "Site (2)",
426             "beta.3" = "Stream (3)",
427             "beta.4" = "4"),
428   drop = TRUE)
429
430 p1 <- ggplot_bb.seco + shared_scale
431 p2 <- ggplot_bb.chuva + shared_scale
432 p3 <- ggplot_ca.seco + shared_scale
433 p4 <- ggplot_ca.chuva + shared_scale
434
435 painel_rios <- (p1 + p2) / (p3 + p4) +
436   plot_layout(guides = "collect") &

```

```

437   theme(legend.position = "right",
438         legend.title = element_blank())
439
440 painel_rios
441
442 ggsave(
443   filename = "res/fig-painel_pt.rios.png",
444   plot = painel_rios,
445   width = 7,
446   height = 8,
447   dpi = 300
448 )

1  #TABELA DE PARTIÇÃO----
2
3  #CRIANDO UMA FUNÇÃO DE EXTRAÇÃO
4
5  extrair_linha_particao <- function(part_obj, sistema, periodo) {
6
7    mapa_componentes <- c(
8      "alpha.1" = " 1 (Habitat)",
9      "alpha.2" = " 2 (Site)",
10     "alpha.3" = " 3 (Stream)",
11     "beta.1"  = " 1 (Between habitats)",
12     "beta.2"  = " 2 (Between sites)",
13     "beta.3"  = " 3 (Between streams)",
14     "gamma"   = " (Total richness)"
15   )
16
17   stat <- part_obj$statistic
18   comp_names <- names(stat)
19
20   df <- data.frame(
21     Sistema      = sistema,
22     Período      = periodo,
23     Componente   = ifelse(comp_names %in% names(mapa_componentes),
24                           mapa_componentes[comp_names], comp_names),
25     Observado    = sprintf("%.2f%%", stat * 100),
26     Esperado     = sprintf("%.2f%%", part_obj$oeocosimu$means * 100),
27     SES          = sprintf("%.2f", part_obj$oeocosimu$z),
28     p_val        = part_obj$oeocosimu$pval,
29     stringsAsFactors = FALSE
30   )

```

```

31
32 df$p_valor <- ifelse(df$p_val <= 0.001, "< 0.001*",
33                       ifelse(df$p_val <= 0.05, sprintf("%.3f*", df$p_val),
34                             sprintf("%.3f", df$p_val)))
35
36 return(df %>% select(-p_val))
37 }
38
39 #PROCESSAMENTO E RUNDOWN DOS OBJETOS ADIPART
40
41 # Rodando o adipart para cada cenário e gerando os objetos corretos
42
43 #pt.total (m_cont)
44 #pt.seco (m_cont_s), pt.chuva (m_cont_c)
45 #bb.seco(A) (m_cont_bb.seco), bb.chuva(B) (m_cont_bb.chuva)
46 #ca.seco(C) (m_cont_ca.seco), ca.chuva(D) (m_cont_ca.chuva)
47
48 part_p_pt.total <- adipart(y = m_cont, x = pt.total, index = "richness", relative = TRUE,
49                            weights = "unif", nsimul = 999)
50
51 set.seed(666)
52 part_p_pt.seco <- adipart(y = m_cont_s, x = pt.seco, index = "richness", relative = TRUE,
53                           weights = "unif", nsimul = 999)
54 part_p_pt.chuva <- adipart(y = m_cont_c, x = pt.chuva, index = "richness", relative = TRUE,
55                            weights = "unif", nsimul = 999)
56
57 part_p_bb.seco <- adipart(y = m_cont_bb.seco, x = bb.seco, index = "richness", relative = TRUE,
58                           weights = "unif", nsimul = 999)
59 part_p_bb.chuva <- adipart(y = m_cont_bb.chuva, x = bb.chuva, index = "richness", relative = TRUE,
60                            weights = "unif", nsimul = 999)
61
62 part_p_ca.seco <- adipart(y = m_cont_ca.seco, x = ca.seco, index = "richness", relative = TRUE,
63                           weights = "unif", nsimul = 999)
64 part_p_ca.chuva <- adipart(y = m_cont_ca.chuva, x = ca.chuva, index = "richness", relative = TRUE,
65                            weights = "unif", nsimul = 999)
66
67 part_p_pt.seco <- readRDS("res/part_p_pt.seco.rds")
68
69
70 part_p_pt.total <- readRDS("res/part.p-pt.total.rds")
71 part_p_pt.seco <- readRDS("res/part.p-pt.seco.rds")
72 part_p_pt.chuva <- readRDS("res/part.p-pt.chuva.rds")
73 part_p_bb.seco <- readRDS("res/part.p-bb.seco.rds")

```

```

74 part_p_bb.chuva <- readRDS("res/part.p-bb.chuva.rds")
75 part_p_ca.seco <- readRDS("res/part.p-ca.seco.rds")
76 part_p_ca.chuva <- readRDS("res/part.p-ca.chuva.rds")
77
78 # CONSOLIDAÇÃO DA TABELA
79
80 tabela_raw <- rbind(
81   extrair_linha_particao(part_p_pt.seco, "Partição Total (PT)", "Seco"),
82   extrair_linha_particao(part_p_pt.chuva, "Partição Total (PT)", "Chuvoso"),
83   extrair_linha_particao(part_p_bb.seco, "Riacho Barro Branco (BB)", "Seco"),
84   extrair_linha_particao(part_p_bb.chuva, "Riacho Barro Branco (BB)", "Chuvoso"),
85   extrair_linha_particao(part_p_ca.seco, "Riacho Caiana (CA)", "Seco"),
86   extrair_linha_particao(part_p_ca.chuva, "Riacho Caiana (CA)", "Chuvoso")
87 )
88
89 # IMPRESSÃO DA TABELA GT
90
91 tabela_final <- tabela_raw %>%
92   gt(groupname_col = "Sistema") %>%
93   cols_label(
94     Período = "Período",
95     Componente = "Componente Hierárquico",
96     Observado = "Observado (%)",
97     Esperado = "Esperado (%)",
98     SES = "SES",
99     p_valor = "p-valor"
100 ) %>%
101   cols_align(
102     align = "center",
103     columns = c("Período", "Observado", "Esperado", "SES", "p_valor")
104   )
105 tabela_final
106
107 # Formatar nomes das colunas diretamente
108 tabela_simples <- tabela_raw
109 colnames(tabela_simples) <- c("Sistema", "Período", "Componente", "Obs (%)", "Esp (%)", "SES")
110
111 # Imprimir no console
112 print(tabela_simples, row.names = FALSE)
113
114 # Exportando dados para Excel
115 library(openxlsx)
116 #write.xlsx(tabela_simples, file = "res/tabela de partição.xlsx", rowNames = FALSE)

```

```

117 wb <- loadWorkbook("res/tabela de partição.xlsx")
118 writeData(wb, sheet = "Sheet 1", x = tabela_simples)
119 saveWorkbook(wb, "res/tabela de partição.xlsx", overwrite = TRUE)
120
121 # Tabela final ajustada no Excel
122 library(readxl)
123 part_gtttable <- read_excel(
124   path = "res/tabela de partição.xlsx",
125   sheet = "tabela_final",
126   range = "A1:G28") #ajustar para o tamanho da tabela final
127 part_gtttable
128 library(gt)
129 part_gtttable <- gt(part_gtttable)
130 part_gtttable <- sub_missing(part_gtttable,
131                               columns = everything(), missing_text = "")
132 part_gtttable
133 gtsave(part_gtttable, "res/gt-part_gtttable_xlsx.html")
134 saveRDS(part_gtttable, "res/gt-part_gtttable_xlsx.rds")

```

4 Discussion

Depth and width showed greater variations across sites as a result from the fact the streams are barred at some points from roads for local traffic and small dams for water storage. Stream width and depth showed greater variations reflecting the range of habitats along the study streams, from small and narrow streams to wider and deeper manmade reservoirs.

Conclusions

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Authorship contribution statement

Authorship of this paper is based on CRediT (2026).

Ellen Gomes da Silva: Data curation, Formal analysis, Investigation, Software, Visualization, Writing – original draft.

Mayara Mirelly da Silva Monteiro: Data curation, Formal analysis, Investigation.

Ludmilla Cavalcanti Antunes Lucena: Supervision, Validation.

Thais Xavier de Melo: Supervision, Validation.

Elvio Sergio Figueredo Medeiros: Conceptualization, Data Curation, Formal Analysis, Funding Acquisition, Investigation, Methodology, Project Administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Ethical approval

Activities inside the REBIO Guaribas were authorized by the Instituto Chico Mendes de Conservação da Biodiversidade (license no. SISBIO-89415-1, SISBIO-89429-2 and SISBIO-89415-2).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, the authors used ChatGPT (OpenAI) to assist with English language editing and improve the clarity and fluency of the writing. No AI was used for data analysis, interpretation, or generation of scientific conclusions. The authors carefully reviewed and revised all AI-generated suggestions and take full responsibility for the accuracy and content of the final manuscript (CNPq 2026).

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Figures and Tables

Figure 1: Location of sampling sites along the Barro Branco Stream (Mamanguape, PB) (Site 1 06°43'06"S and 35°10'54"W; Site 2 06°42'38"S and 35°10'38"W; Site 3 06°43'06"S and 35°10'54"W; Site 4 06°41'46"S and 35°10'36"W; Site 5 06°40'56"S and 35°10'27"; Site 6 06°40'18"S and 35°10'34"W).

Figure 2: Principal Component Analysis (PCA) biplot of the sampling units (SUs) and environmental variables in the Barro Branco and Caiana streams during the study period, categorized by stream. Points represent the SUs, whereas vectors represent the environmental variables associated with habitat structure. Ellipses represent 95% confidence regions for the distribution of the groups of points, calculated based on the t-distribution. Only variables that contributed at least 20% to the axes are shown.

Figure 3: Principal Component Analysis (PCA) biplot of the sampling units (SUs) and environmental variables in the Barro Branco and Caiana streams during the study period, categorized by season. Points represent the SUs, whereas vectors represent the environmental variables associated with habitat structure. Ellipses represent 95% confidence regions for the distribution of the groups of points, calculated based on the t-distribution. Only variables that contributed at least 20% to the axes are shown.

Figure 4: Individual-based extrapolation and rarefaction curves for the richness of resting stages of zooplanktonic taxa in each study stream. Shaded areas indicate 95% confidence intervals.

Figure 5: Classification analysis (UPGMA using Bray-Curtis dissimilarity) of wetland macroinvertebrate samples obtained in a. permanent and b. intermitent wetlands. The first number of the sample label indicates the wetland. The second number indicates the habitat type (1=hydrophytes; 2=reed-like emergent plants; and 3=leafy emergent plants). The third number indicates the individual sweeps realized in each habitat (1-3)UPGMA Hierarchical flexible beta clustering ($\beta = -0.25$) based on Bray-Curtis distance of (a) macroinvertebrates and (b) Chironomidae in the Paraíba River during the 2015 wet season. Tag codes indicate different sections of the river: U = upper segment, M = middle segment, and T = Taperoá River segment. R = reaches (1-3), S = sites (1-3), and Hab = habitat types (1-3). Percent chaining <4%.

Figure 6: Observed and expected diversity, partitioned into one alpha and three beta components, expressed as percent of total species richness. Numbers indicate the proportion of randomized samples containing more species than the observed sample for each partition. 1: subsample species richness, 1: beta diversity among sub-samples; 2: beta diversity among pools or habitats within pool, and 3: beta diversity among river reaches.

Figure 7: Observed and expected diversity, partitioned into one alpha and three beta components, expressed as percent of total species richness. Numbers indicate the proportion of randomized samples containing more species than the observed sample for each partition. 1:

subsample species richness, 1: beta diversity among sub-samples; 2: beta diversity among pools or habitats within pool, and 3: beta diversity among river reaches.

Table 1: Environmental variables averaged for sampling sites (minimum-maximum) for the Barro Branco stream (Mamanguape, PB).

Table 2: Nutrient concentration (g L⁻¹) averaged for sampling sites (minimum-maximum) for the Barro Branco stream (Mamanguape, PB).

Table 3: List of species and counts of zooplankton collected during the 2024 hydrological cycle.

Table 4: Additive partitioning of zooplankton species richness (γ -diversity) into hierarchical alpha (α) and beta (β) diversity components across sampling levels (expressed in percentage values). Shown are the observed diversity values, mean diversity values expected under the null model, standardized effect sizes (SES), and statistical significance (P -value). Numbers in ‘ P -value’ column indicate the proportion of randomized samples containing more species than the observed sample for each partition. α_1 (Habitat): Average species richness per habitat. β_1 (Habitat): Diversity turnover among habitats within a site. β_2 (Site): Diversity turnover among sampling sites within a stream. β_3 (Stream): Diversity turnover between streams.

Figures

Figure 1

Figure 2

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Tables

Table 1

Table 2

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Table 4

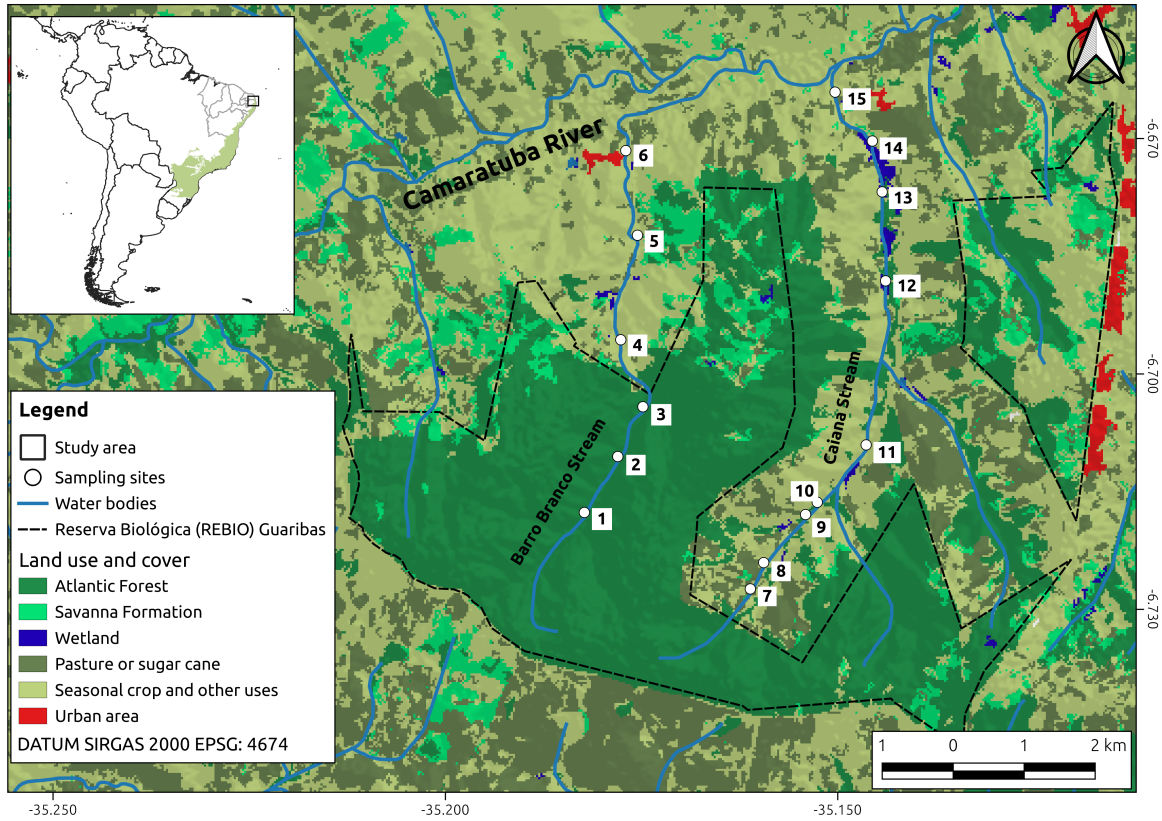


Figure 1: Study area

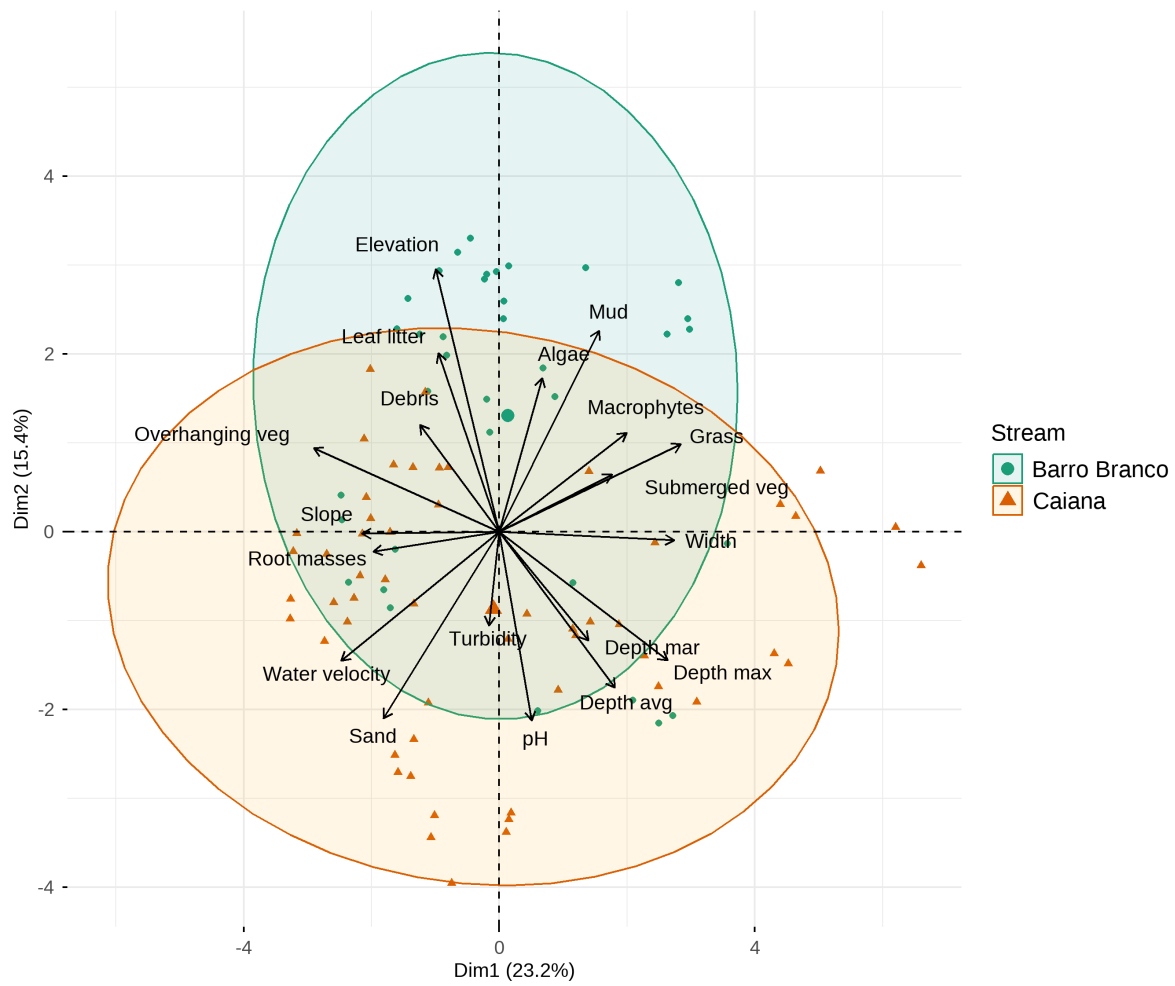


Figure 2: Principal Component Analysis, streams

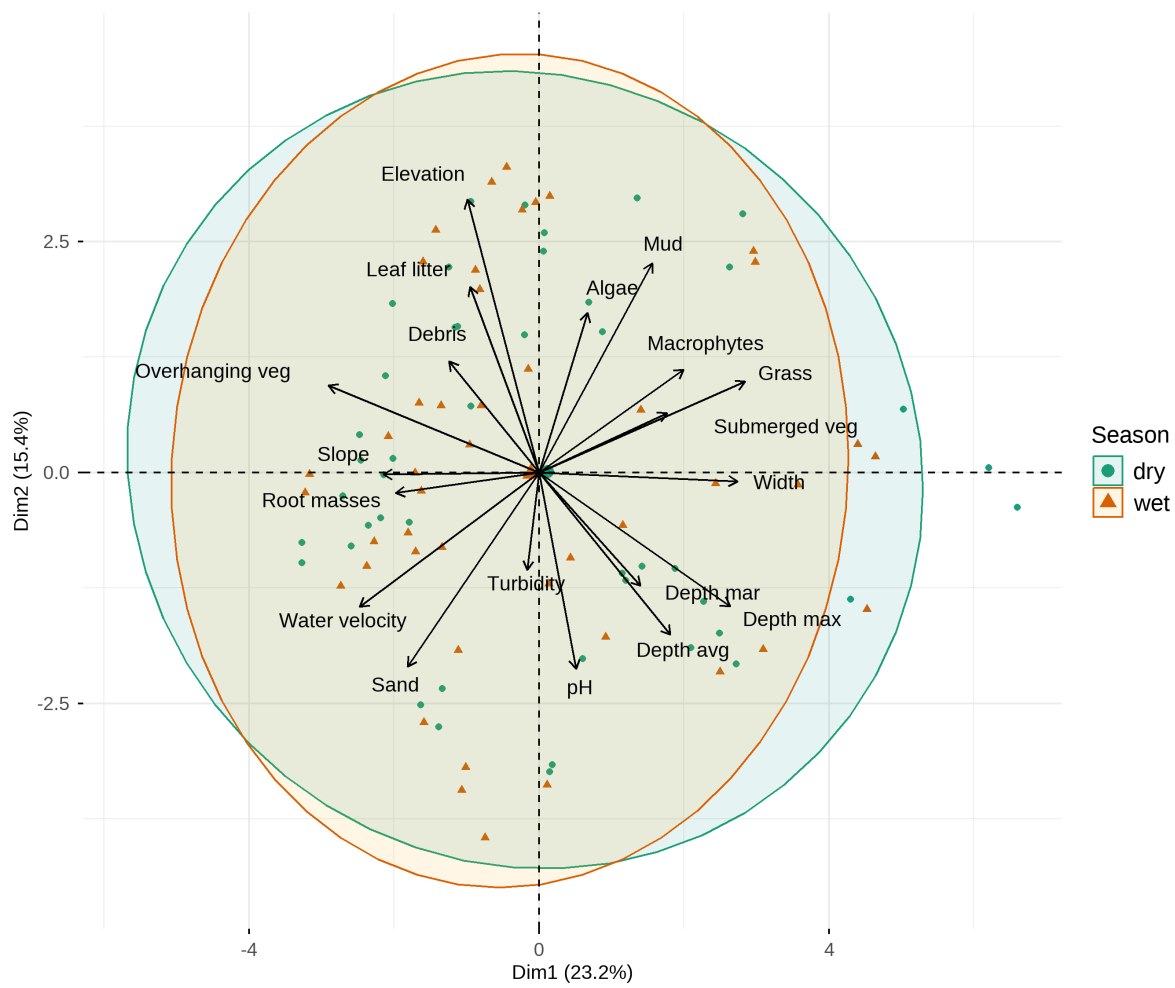


Figure 3: Principal Component Analysis, seasons

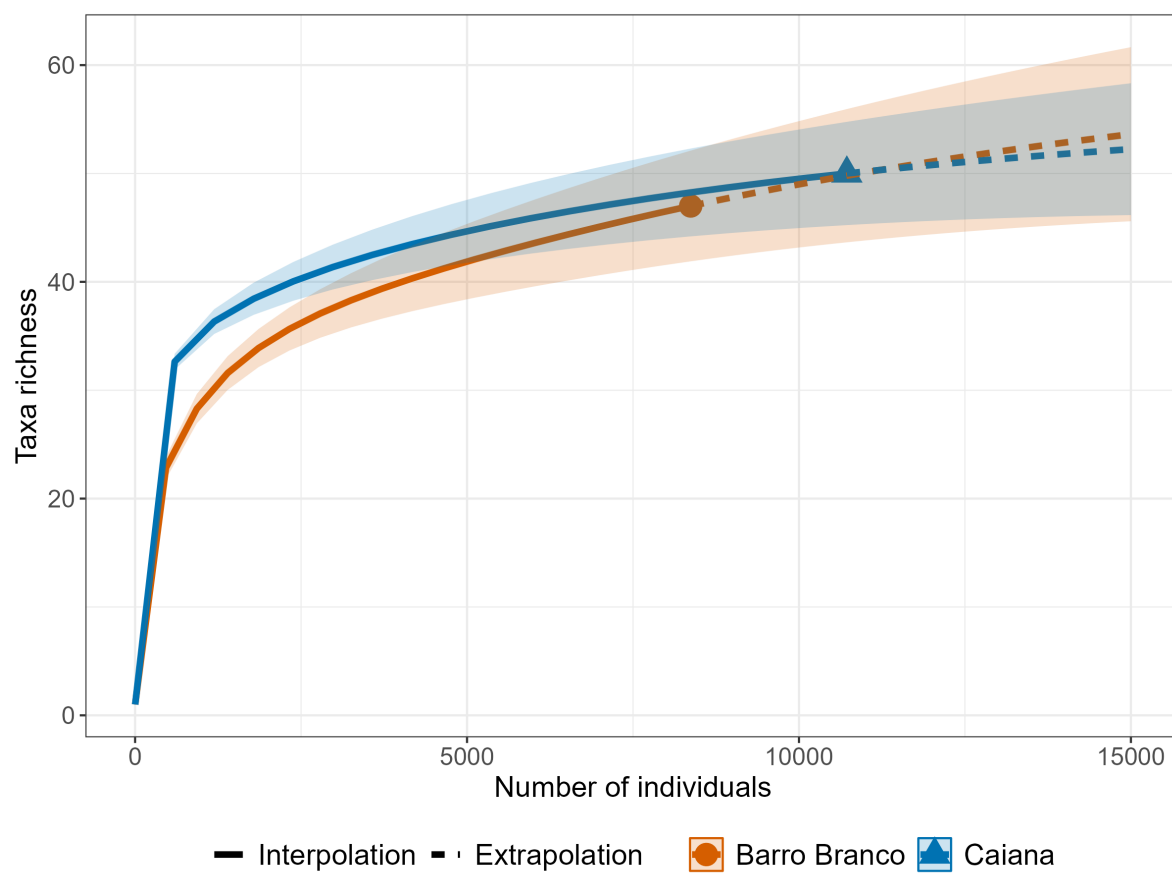


Figure 4: Accumulation curves

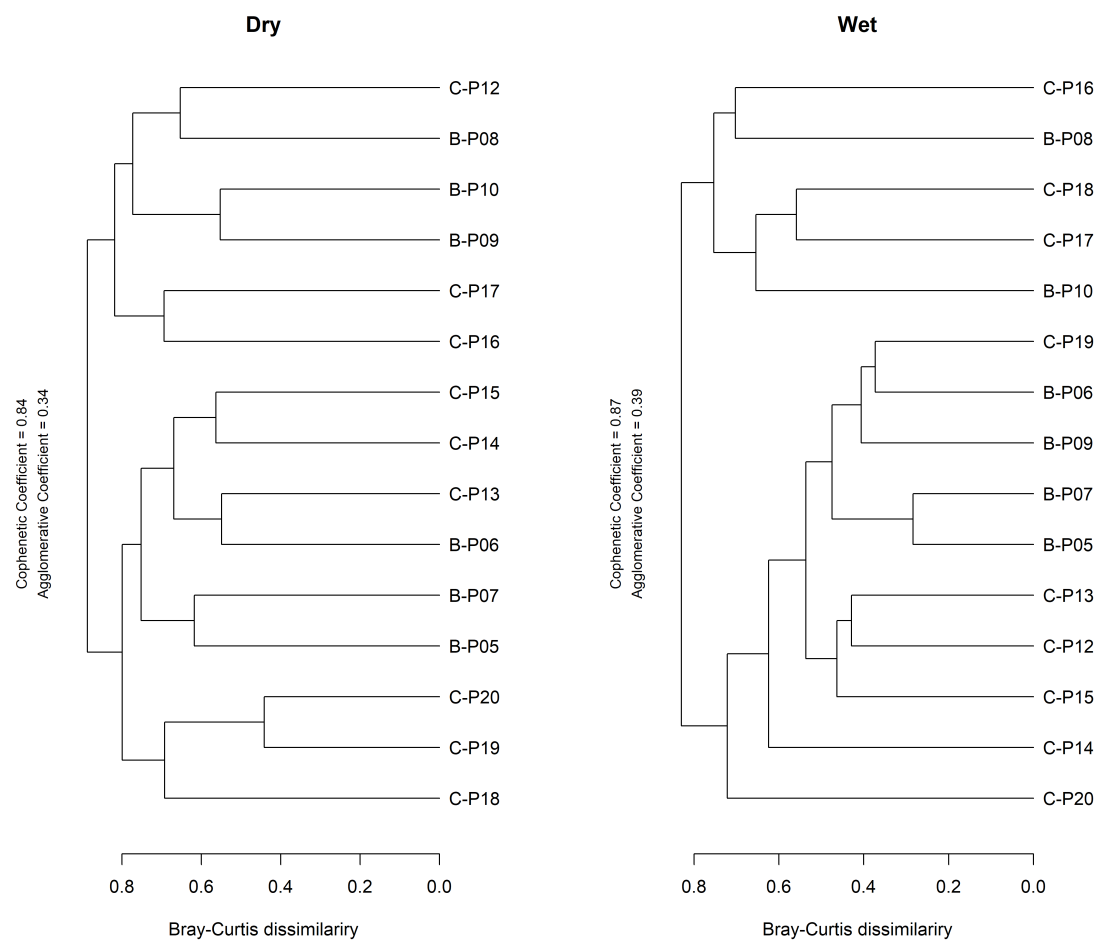


Figure 5: UPGMA Classification

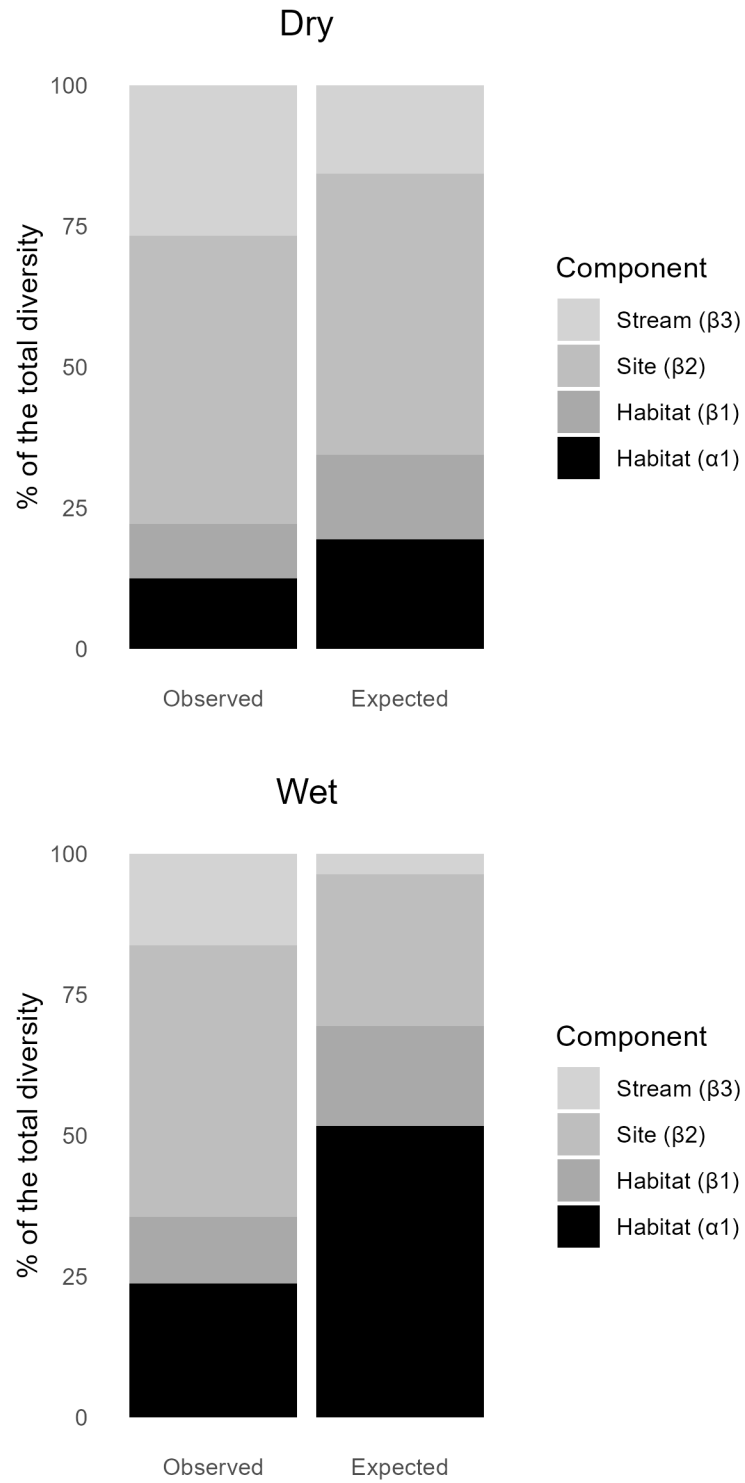


Figure 6: Partitioning plots season

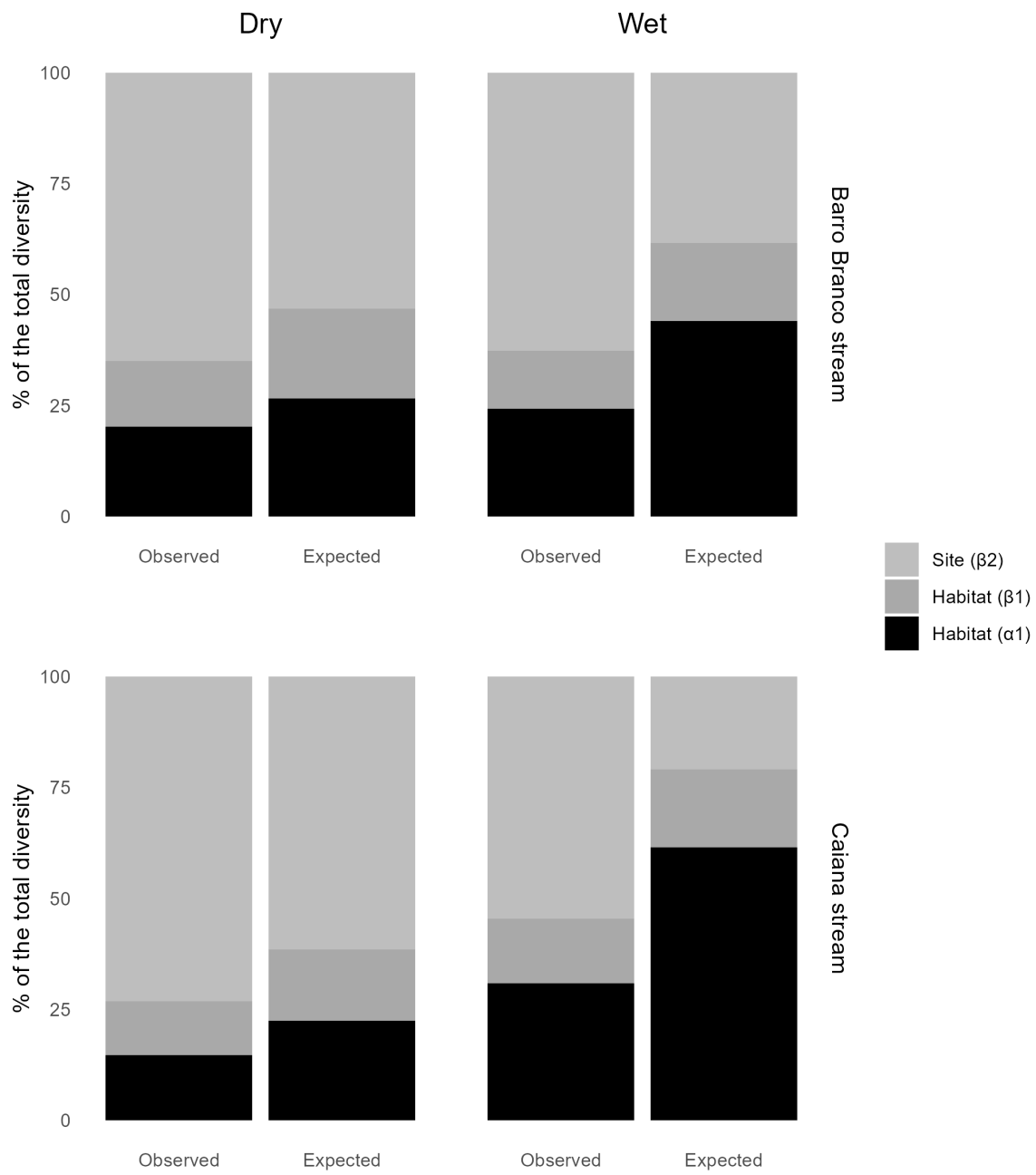


Figure 7: Partitioning plots stream

Table 1: Environment data GT table.

...1	Barro Branco stream	...3	...4	...5
	Site 1	Site 2	Site 3	Site 4
Water quality				
Temperature (°C)*	25.8 (24.9-27.3)	26.5 (25.3-27.5)	25.2 (24.9-25.5)	27.8 (27.3-28.3)
Dissolved oxygen (mg/L)*	8.1 (7.8-8.5)	7.5 (7.2-7.8)	7.9 (7.7-8)	7.5 (6.5-8.5)
Water velocity (m/s)	0.073 (0.025-0.161)	0.086 (0.035-0.24)	0.135 (0.022-0.207)	0.036 (0.01-0.1)
Conductivity (µS/cm)	137.5 (119.2-164.4)	148 (134.1-156.8)	137.5 (98.3-160.2)	154.2 (141.5-166.9)
Turbidity (NTU)	5 (2-10.7)	4.9 (2.1-14.1)	4.9 (1.5-9.2)	5.7 (4-6.4)
pH	5.9 (4.7-7.5)	5.7 (4.6-7.7)	5.1 (4.5-6)	5.5 (5-6)
Morphology				
Elevation (m a.s.l.)	107.1 (107.1-107.1)	112.8 (112.8-112.8)	89 (89-89)	89.1 (89.1-89.1)
Slope	1.4 (1-2)	2.7 (2.5-3)	2.4 (1.5-3)	1.5 (1-2)
Average depth (cm)	12.8 (9-14)	23.7 (16.3-31.7)	23.8 (16-41)	43.3 (28-58)
Margin depth (cm)	11.8 (3-16)	21.7 (8-30)	18.8 (7-42)	48.3 (16-80)
Maximum depth (cm)	15.5 (13-17)	32.7 (27-40)	29.3 (20-47)	50.8 (35-66)
Width (m)	6.2 (3.6-14.1)	4.3 (0.7-9.2)	2 (1.1-3.2)	7.6 (2-16)
Substrate composition (%)				
Mud	100 (100-100)	100 (100-100)	95 (90-100)	74.2 (5-100)
Sand	0	0	5 (0-10)	25.8 (0-95)
Gravel	0	0	0	0
Rocks	0	0	0	0
Habitat structures (%)				
Aquatic macrophytes	5.8 (5-10)	10 (0-30)	0 (0-0)	17.9 (0-60)
Algae	2.5 (0-5)	0.2 (0-1)	5.5 (0-21)	19.2 (15-25)
Submerged vegetation	7.5 (5-15)	8.3 (5-15)	5.8 (0-10)	5.8 (0-10)
Overhanging vegetation	68.3 (40-90)	70.8 (50-85)	76.7 (60-90)	31.7 (25-40)
Littoral grass	3.3 (0-10)	8.3 (5-15)	6 (0-15)	27.1 (15-40)
Leaf litter	7.5 (5-10)	11 (1-25)	15.8 (5-30)	18.3 (10-30)
Root masses	5 (0-15)	10.8 (5-15)	12.5 (5-20)	4.2 (0-10)
Wood debris	30.8 (20-45)	12.7 (5-20)	17.7 (5-30)	21.7 (5-30)

Table 2: Nutrient concentrations at the sampling sites.

...1	Barro Branco stream	...3	...4	...5	.
	Site 1	Site 2	Site 3	Site 4	S
Ammonia ($\mu\text{g L}^{-1}$)	54.8 (39.3-64.3)	58.9 (47.5-77)	72 (33.4-115.7)	81.6 (68.9-93)	8
Nitrite ($\mu\text{g L}^{-1}$)	0	0	0	0	0
Nitrate ($\mu\text{g L}^{-1}$)	6.0 (0-9.7)	8.8 (6.9-12.6)	5.5 (0-12.6)	18.8 (11.1-31.1)	1
Orthophosphate ($\mu\text{g L}^{-1}$)	0	0	1.0 (0-3)	0	0
Phosphorus ($\mu\text{g L}^{-1}$)	0	4.6 (0.3-10.3)	50.1 (0-150.3)	9 (0-27)	1

Apendices

Non-used figures and tables

 Non-used codes

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1 table(1:10)
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Table 3: Checklist of taxa GT table.

...1	...2	Barro Branco stream	...4	Caian
ROTIFERA		Dry	Wet	Dry
DIGONONTA		N= 1529	N= 6840	N= 2
Bdelloidea	Bdelloidea	994	1930	835
Philodinidae				
Dissotrocha aculeata (Ehrenberg, 1832)	Dis.aculeata	0	0	2
Dicranophoridae				
Dicranophoroides sp.	Dicranop.sp	0	0	2
Notommatidae				
Cephalodella sp.	Cephalodella.sp	2	8	8
MONOGONONTA				
FLOSCULARIACEA				
Lecanidae				
Lecane aculeata (Jakubski, 1912)	Lec.aculeata	1	0	0
Lecane bulla (Gosse, 1851)	Lec.bulla	60	9	20
Lecane cornuta (Müller, 1786)	Lec.cornuta	4	4	5
Lecane crepida Haring, 1914	Lec.crepida	13	1	2
Lecane curvicornis (Murray, 1913)	Lec.curvicornis	1	5	6
Lecane furcata (Murray, 1913)	Lec.furcata	1	0	0
Lecane leontina (Turner, 1892)	Lec.leontina	3	83	7
Lecane lunaris (Ehrenberg, 1832)	Lec.lunaris	7	5	4
Lecane luna (Müller, 1776)	Lec.luna	13	4	8
Lecane monostyla (Daday, 1897):	Lec.monostyla	1	0	1
Lecane papuana (Murray, 1913)	Lec.papuana	2	0	3
Lecane quadridentata (Ehrenberg, 1830)	Lec.quadridentata	2	0	0
Lecane sp.1	Lecane.sp1	1	0	1
Lecane sp.2	Lecane.sp2	0	0	3
Lecane sp.3	Lecane.sp3	1	0	0
Brachionidae				
Brachionus patulus Müller, 1786	Bra.patulus	3	3	0
Macrochaetus sp.	Macrochaetus.sp	2	25	2
Platytias quadricornis (Ehrenberg, 1832)	Pla.quadricornis	6	176	2
Lepadellidae				
Lepadella ovalis (Müller, 1786)	Lep.ovalis	17	5	3
Lepadella patella (Müller, 1773)	Lep.patella	2	3	0
Lepadella imbricata Haring, 1914	Lep.imbricata	0	0	1
Lepadella sp.	Lepadella.sp	0	0	1
Euchlanidae				
Euchlanis dilatata Ehrenberg, 1832	Euc.dilatata	0	0	1
Synchaetidae				
Polyarthra bicerca	Pol.bicerca	0	1	36
Polyarthra dolichoptera Idelson, 1925	Pol.dolichoptera	2	0	0
Polyarthra vulgaris	Pol.vulgaris	0	1	145
Polyarthra sp.	Polyarthra.sp	8	0	0
Colurellidae				
Colurella sp.	Colurella.sp	8	0	0
Mytilinidae				
Mytilina crassipes (Lucks, 1912)	Myt.crassipes	0	0	1
Mytilina bisulcata (Lucks, 1912)	Myt.bisulcata	1	0	0
Mytilina	Myt.	1	0	1

Table 4: Partition data GT table.

Partition	Season	Component	Diversity...4	Diversity...5	SES	P-value
Partition	Season	Component	Observed (%)	Expected (%)	SES	P-value
Barro Branco stream	Dry	1 (Habitat)	12.45	19.43	-22.13	< 0.00
		(Total richness)	100.00	100.00	0.00	1.000
		1 (Between habitats)	9.73	15.03	-8.93	< 0.00
		2 (Between sites)	51.09	49.81	0.64	0.519
		3 (Between streams)	26.72	15.73	5.57	< 0.00
	Wet	1 (Habitat)	23.72	51.71	-45.44	< 0.00
		(Total richness)	100.00	100.00	0.00	1.000
		1 (Between habitats)	11.78	17.68	-7.03	< 0.00
		2 (Between sites)	48.25	27.02	13.41	< 0.00
		3 (Between streams)	16.25	3.59	9.48	< 0.00
	Dry	1 (Habitat)	20.23	26.63	-11.17	< 0.00
		(Total richness)	100.00	100.00	0.00	1.000
		1 (Between habitats)	14.81	20.17	-4.53	< 0.00
		2 (Between sites)	64.96	53.19	8.94	< 0.00
		1 (Habitat)	24.24	43.98	-20.41	< 0.00
Caiana stream	Wet	(Total richness)	100.00	100.00	0.00	1.000
		1 (Between habitats)	13.13	17.62	-3.13	< 0.00
		2 (Between sites)	62.63	38.41	13.37	< 0.00
		1 (Habitat)	14.73	22.48	-15.49	< 0.00
		(Total richness)	100.00	100.00	0.00	1.000
	Dry	1 (Between habitats)	12.08	15.97	-4.18	0.003*
		2 (Between sites)	73.19	61.55	10.17	< 0.00
		1 (Habitat)	30.83	61.48	-34.59	< 0.00
		(Total richness)	100.00	100.00	0.00	1.000
		1 (Between habitats)	14.60	17.64	-2.85	0.007*
	Wet	2 (Between sites)	54.58	20.88	26.58	< 0.00